

Knowledge Processing

Knowledge Organisation

Heterogeneity in Knowledge

Knowledge Graph Exploitation

Gábor Bella, Yannis Haralambous

Slides and Lab Material:

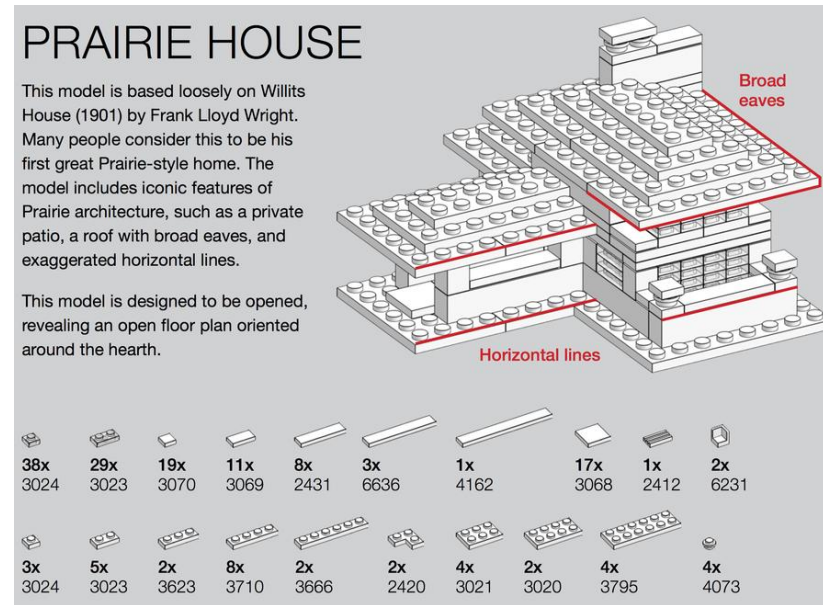
<https://github.com/gbella/NLP>, under Knowledge_Processing

Knowledge Processing Knowledge Organisation

Gábor Bella, Yannis Haralambous

What is Knowledge Organisation?

- Over millennia, human activity has produced enormous amounts of knowledge.
- Such knowledge needs to be structured.
- Knowledge representation (KR) is a field of AI that deals with the fundamentals of formally expressing knowledge: it is a bottom-up process.
- Knowledge organisation (KO) complements KR with a top-down approach.
- KO is as old as civilisation: the first libraries already had KO systems.



Knowledge
Organisation



Knowledge
Representation

Examples of KO: Dictionaries, Thesauri, Terminologies

- **Dictionaries** collect knowledge about language, more precisely about the lexicon.
- **Lexemes** are presented in their canonical forms (e.g. infinitive for verbs) and in a standard (e.g. alphabetical) order. Phenomena of lexical heterogeneity, such as homonymy, polysemy, or synonymy, are addressed.
- A **monolingual dictionary** typically provides a list of possible meanings for a lexeme, examples of use, etymology, etc.
- A **thesaurus** is a more formal monolingual dictionary that also provides relationships across lexemes and their meanings (broader/narrower, part-of, antonymy, synonymy, etc.).
- A **bilingual dictionary** provides possible translations between lexemes of two languages.
- A **domain terminology** (monolingual or multilingual) concentrates on the lexemes (terms) of a given knowledge domain, as opposed to general language. Domain terminologies aim to be exhaustive and can be larger than dictionaries concentrating on the general lexicon.



The first known, Sumerian-Akkadian dictionary from about 2000 BC.

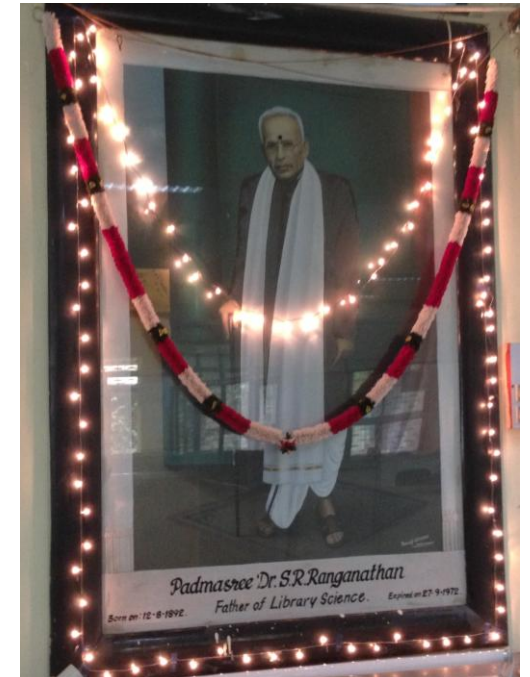
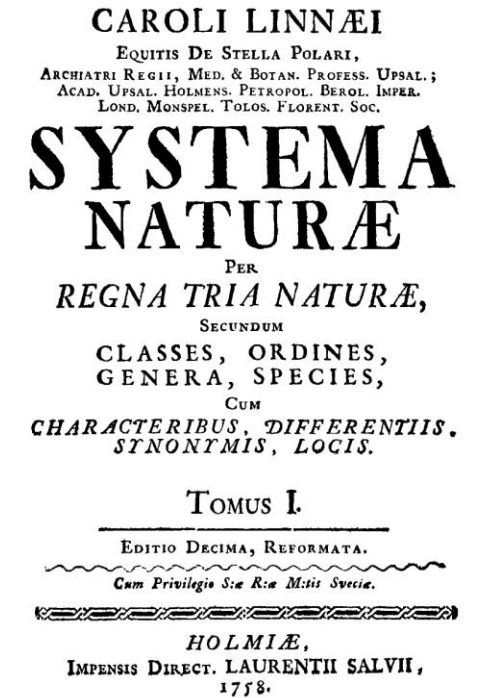
Examples of KO: Encyclopedias

- Dictionaries catalogue lexemes, concentrating mostly on common nouns, verbs, adjectives, adverbs.
- Proper nouns (people, locations, events, etc.) are excluded from dictionaries, as their number is orders of magnitude larger.
- Another difference is the rate of change: while everything in the world keeps changing, including human language and knowledge, facts about entities of the world (objects, people, events) evolve at a much higher pace.
- Such facts are rather collected in encyclopedias.
- Encyclopedias use natural language (and sometimes images) to describe their entries.



Examples of KO: Classifications

- [Linné, 1735: Systema Naturæ](#), the first scientific classification of the vegetal, animal, and mineral kingdoms.
- [Library classifications](#): probably the most ancient classification systems (e.g. library of Alexandria).
 - Dewey Decimal Classification, 1876 (USA).
 - Universal Decimal Classification, 1895 (Belgium).
 - [Colon Classification](#), 1933, by Ranganathan, the greatest theorician of library classification. It is a [faceted classification](#) where each level of the hierarchy (facet) has a precise meaning.
- [International Classification of Diseases](#) (Bertillon, 1893): one of the most important norms in healthcare today, its revisions 9, 10, and 11 are currently in use around the world.
- Such classifications remain used today in various branches of human activity, available as electronic resources.



Organisation of Formal Knowledge

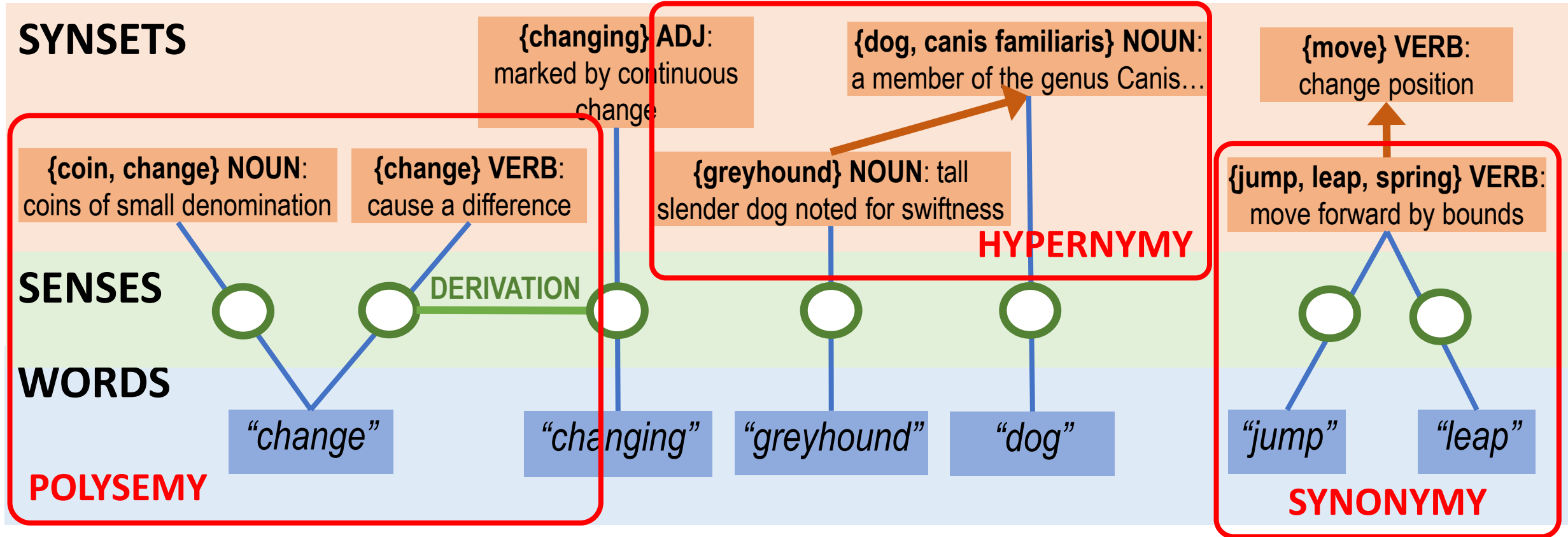
- **Classic human KO approaches remain dominant in symbolic AI:**
the direct mapping between human-interpretable and computer-interpretable knowledge is key to explainability.
- You already saw such examples in KR:
 - Lexical resources: dictionaries, thesauri, terminologies => wordnets, domain term bases;
 - Classifications: categories and hierarchy => ontologies, taxonomies;
 - Encyclopedias: relevant objects (persons, places, events, etc.) and facts about the world => subject-predicate-object triples in RDF, as in DBPedia.
- A **knowledge graph** is able to combine all such knowledge.



Wordnets: Knowledge Graphs on General Language

- *WordNet* = the English WordNet designed at Princeton;
wordnet = similarly-structured knowledge graph in any language.
- A wordnet describes the meanings of common nouns, verbs, adjectives, and adverbs of a language, as well as a few proper nouns.
- They are standard reference resources for word meanings in NLP applications.
- The English Princeton wordnet is of very high quality, rich (>100k entries), small, and fast to use. <http://wordnet.princeton.edu>
- It is likely that a wordnet in your language exists, but they are of varying quality and size. Some examples:
 - large and high quality: English, German;
 - large and low quality: French, Indonesian, Arabic;
 - small and high quality: Scottish Gaelic, Setswana.
- Downloadable wordnet-like lexicons for >2,000 languages: <http://ukc.datascientia.eu>

How the Wordnet Deals with Lexical Heterogeneity



Domain Classifications

International Classification of Diseases, v10:
<https://icd.who.int/browse10/2019/en>

North American Industry Classification System:
<https://www.census.gov/naics/?58967?yearbck=2022>

Dewey Decimal Classification:
https://en.wikipedia.org/wiki/List_of_Dewey_Decimal_classes

CHAPTER 9: Diseases of the circulatory system (I00-I99)

Cerebrovascular diseases (I60-I69)

I69 Sequelae of cerebrovascular disease

I69.0 Sequelae of nontraumatic subarachnoid hemorrhage

I69.00 Unspecified sequelae of nontraumatic subarachnoid hemorrhage

I69.01 Cognitive deficits following nontraumatic subarachnoid hemorrhage

I69.010 Attention and concentration deficit following nontraumatic subarachnoid hemorrhage

I69.011 Memory deficit following nontraumatic subarachnoid hemorrhage

I69.012 Visuospatial deficit and spatial neglect following nontraumatic subarachnoid hemorrhage

I69.013 Psychomotor deficit following nontraumatic subarachnoid hemorrhage

I69.014 Frontal lobe and executive function deficit following nontraumatic subarachnoid hemorrhage

I69.015 Cognitive social or emotional deficit following nontraumatic subarachnoid hemorrhage

I69.018 Other symptoms and signs involving cognitive functions following nontraumatic subarachnoid hemorrhage

I69.019 Unspecified symptoms and signs involving cognitive functions following nontraumatic subarachnoid hemorrhage

I69.02 Speech and language deficits following nontraumatic subarachnoid hemorrhage

I69.020 Aphasia following nontraumatic subarachnoid hemorrhage

I69.021 Dysphasia following nontraumatic subarachnoid hemorrhage

I69.022 Dysarthria following nontraumatic subarachnoid hemorrhage

I69.023 Fluency disorder following nontraumatic subarachnoid hemorrhage

I69.028 Other speech and language deficits following nontraumatic subarachnoid hemorrhage

I69.03 Monoplegia of upper limb following nontraumatic subarachnoid hemorrhage

I69.031 Monoplegia of upper limb following nontraumatic subarachnoid hemorrhage affecting right dominant side

I69.032 Monoplegia of upper limb following nontraumatic subarachnoid hemorrhage affecting left dominant side

I69.033 Monoplegia of upper limb following nontraumatic subarachnoid hemorrhage affecting right non-dominant side

I69.034 Monoplegia of upper limb following nontraumatic subarachnoid hemorrhage affecting left non-dominant side

I69.039 Monoplegia of upper limb following nontraumatic subarachnoid hemorrhage affecting unspecified side

I69.04 Monoplegia of lower limb following nontraumatic subarachnoid hemorrhage

I69.041 Monoplegia of lower limb following nontraumatic subarachnoid hemorrhage affecting right dominant side

I69.042 Monoplegia of lower limb following nontraumatic subarachnoid hemorrhage affecting left dominant side

I69.043 Monoplegia of lower limb following nontraumatic subarachnoid hemorrhage affecting right non-dominant side

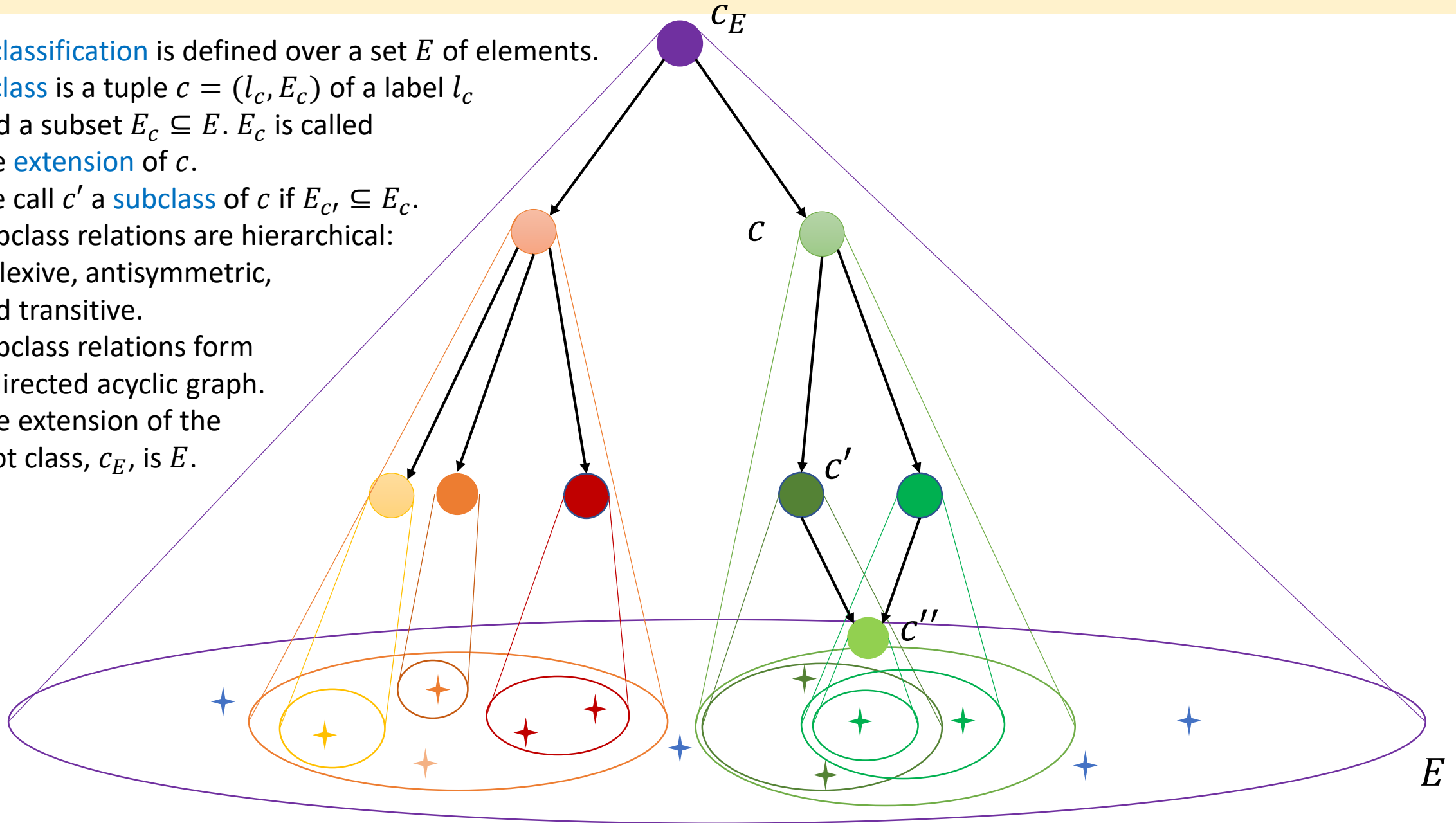
I69.044 Monoplegia of lower limb following nontraumatic subarachnoid hemorrhage affecting left non-dominant side

Sector	Definition
11	Agriculture, Forestry, Fishing and Hunting
21	Mining, Quarrying, and Oil and Gas Extraction
22	Utilities
23	Construction
31-33	Manufacturing
42	Wholesale Trade
44-45	Retail Trade
48-49	Transportation and Warehousing
51	Information
52	Finance and Insurance
53	Real Estate and Rental and Leasing
54	Professional, Scientific, and Technical Services
55	Management of Companies and Enterprises
56	Administrative and Support and Waste Management and Remediation Services
61	Educational Services
62	Health Care and Social Assistance
71	Arts, Entertainment, and Recreation
72	Accommodation and Food Services
81	Other Services (except Public Administration)
92	Public Administration

000 GENERAL WORKS 020 Library and Information Science 030 General Encyclopedias 050 General Periodicals 060 General Organizations	600 TECHNOLOGY 610 Medical Sciences 620 Engineering 630 Agriculture 640 Domestic Sciences 650 Business and Management 660 Chemical Technology 670 Manufacturers 690 Building Construction
100 PHILOSOPHY 110 Metaphysics 130 Psychology, occultism 140 Philosophy 150 Psychology 160 Logic	700 THE ARTS 710 Landscape and Civic Art 720 Architecture 730 Sculpture, Plastics 740 Drawing, Decorative Arts 750 Painting 760 Prints and Print Making 770 Photography 780 Music 790 Recreation, Performing Arts
200 RELIGION 220 The Bible 230 Christian Doctrine 290 Comparative and Other Religions	800 LITERATURE 810 American Literature 820 English Literature 830 German Literature 840 French Literature 850 Italian, Rumanian Literature 860 Spanish, Portuguese Literature 870 Latin and Other Italic Literatures 880 Classical and Modern Greek Literature 890 Other Literature
300 SOCIAL SCIENCES 310 Statistics 320 Political Science 330 Economics 340 Law 350 Public Administration 360 Social Welfare 370 Education 380 Public Service 390 Customs and Folklore	900 HISTORY, GEOGRAPHY 910 Geography, Travel 920 Biography, Genealogy 930 Ancient History 940 Europe 950 Asia 960 Africa 970 North America 980 South America 990 Pacific Ocean Islands 991 Indonesia 993 New Zealand and Melanesia 994 Australia 995 New Guinea (Papua) 996 Polynesia 997 Atlantic Ocean Islands 998 Arctic Region 999 Antarctic Regions
400 LANGUAGE 410 Comparative Linguistics 420 English and Anglo Saxon 430 German Language 440 French 450 Italian, Rumanian 460 Spanish, Portuguese 470 Latin and Other Italic Languages 480 Classical and Modern Greek 490 Other Languages	
500 SCIENCE 510 Mathematics 520 Astronomy 530 Physics 540 Chemistry 550 Earth Sciences 560 Paleontology 570 Life Sciences 580 Botanical Sciences 590 Zoological Sciences	

Classification Hierarchies

- A **classification** is defined over a set E of elements.
- A **class** is a tuple $c = (l_c, E_c)$ of a label l_c and a subset $E_c \subseteq E$. E_c is called the **extension** of c .
- We call c' a **subclass** of c if $E_{c'} \subseteq E_c$.
- Subclass relations are hierarchical: reflexive, antisymmetric, and transitive.
- Subclass relations form a directed acyclic graph.
- The extension of the root class, c_E , is E .

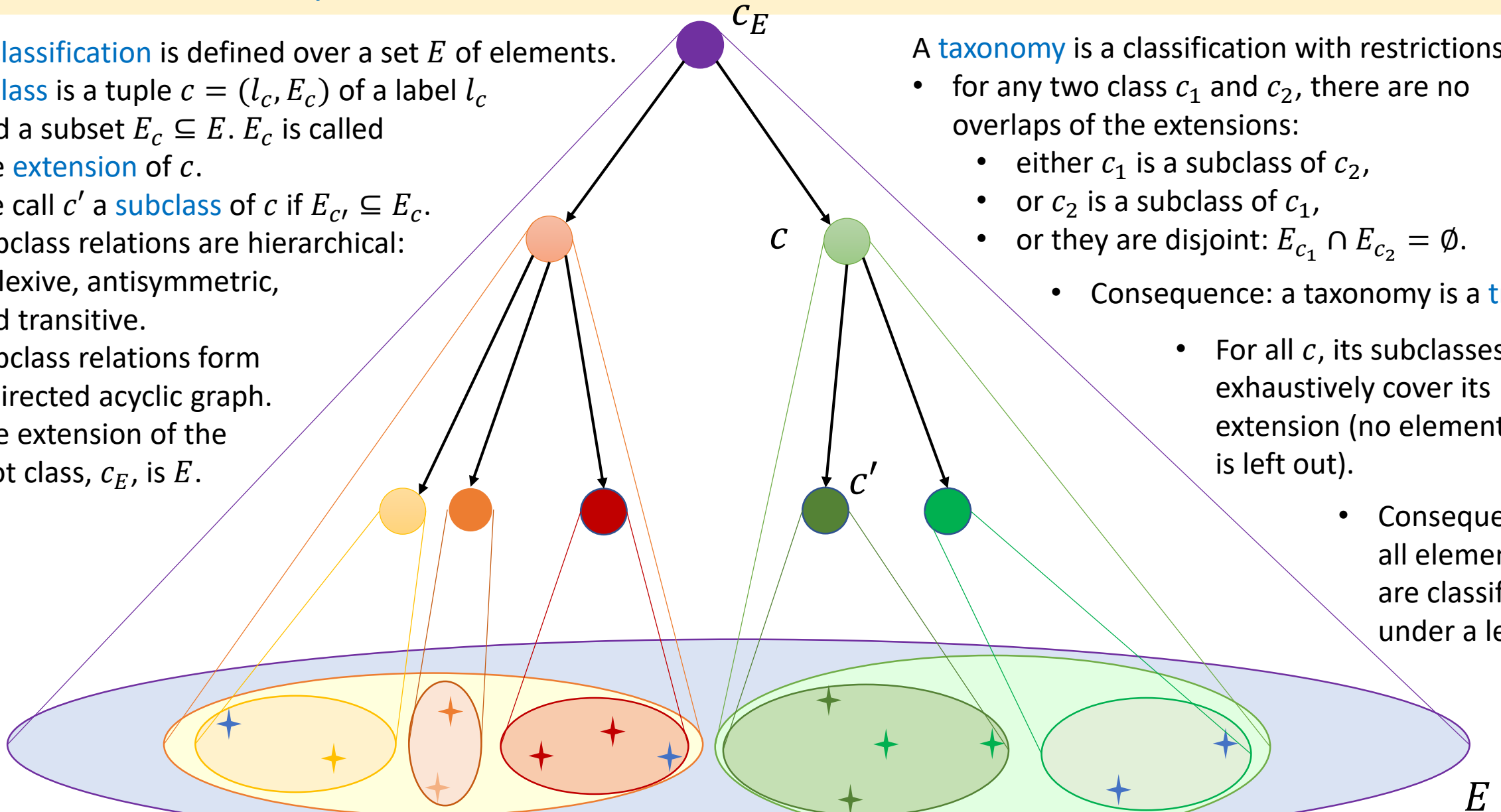


(Hierarchical) Taxonomies

- A **classification** is defined over a set E of elements.
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- Subclass relations are hierarchical: reflexive, antisymmetric, and transitive.
- Subclass relations form a directed acyclic graph.
- The extension of the root class, c_E , is E .

A **taxonomy** is a classification with restrictions:

- for any two class c_1 and c_2 , there are no overlaps of the extensions:
 - either c_1 is a subclass of c_2 ,
 - or c_2 is a subclass of c_1 ,
 - or they are disjoint: $E_{c_1} \cap E_{c_2} = \emptyset$.
- Consequence: a taxonomy is a **tree**.
- For all c , its subclasses exhaustively cover its extension (no element is left out).
- Consequence: all elements are classified under a leaf.



Layers of Knowledge: by Depth

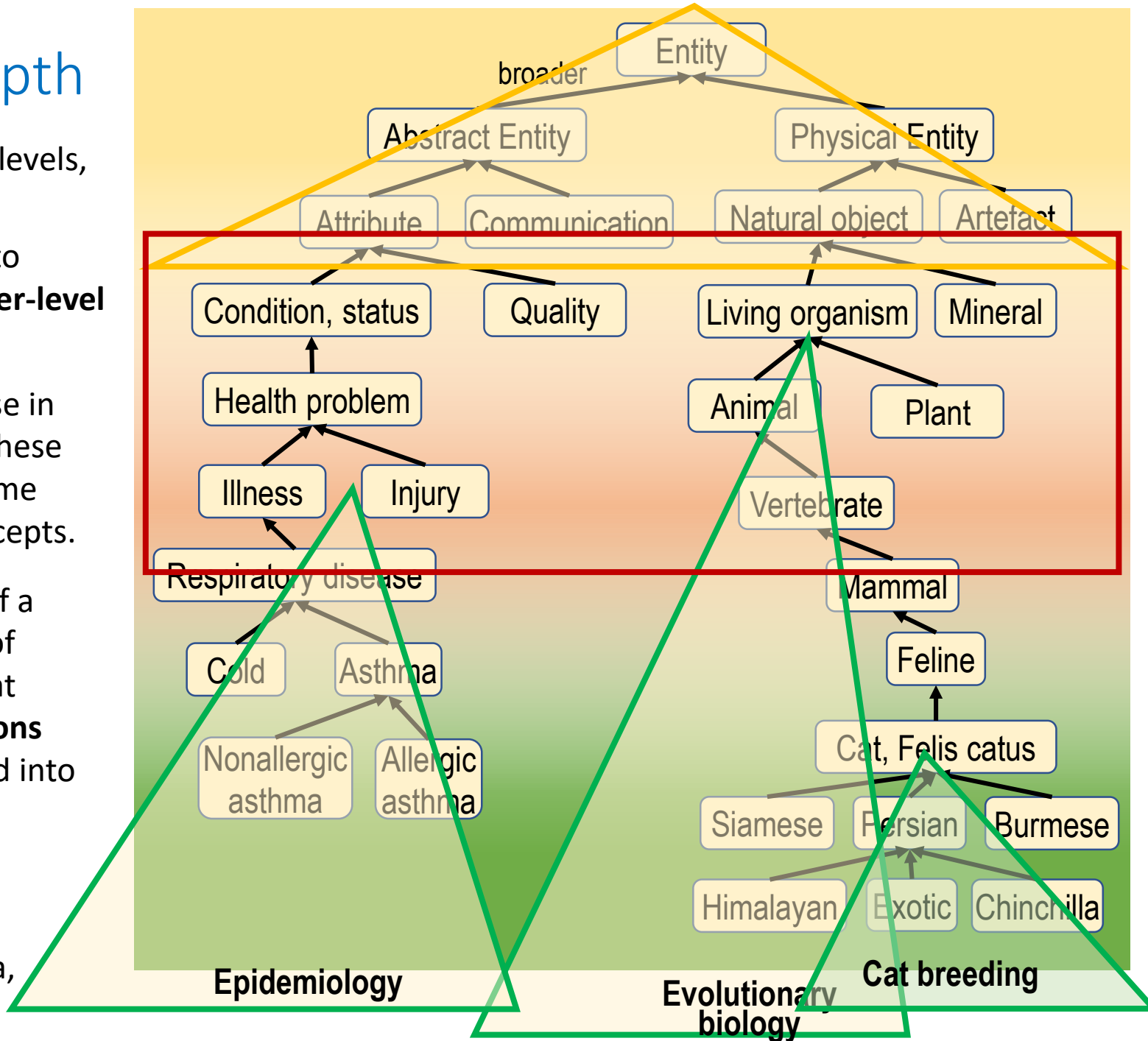
Phenomena can be conceptualised on different levels,
with different degrees of detail.

Top-level concepts are the broadest, they tend to belong to philosophy. They are provided by **upper-level ontologies**.

Basic-level concepts are everyday notions we use in general communication, e.g. “illness” or “cat”. These are the concepts we learn the earliest in life. Some resources, such as **wordnets**, describe such concepts.

Domain-level concepts are used by specialists of a domain. They aim to describe a small segment of knowledge as accurately as possible, and in great detail. They are provided by **domain classifications and ontologies**. Domains are themselves divided into subdomains and application areas.

There is no clear-cut distinction between these layers, and even across domains. In communication, which includes all forms of data, we tend to mix general and domain language. 

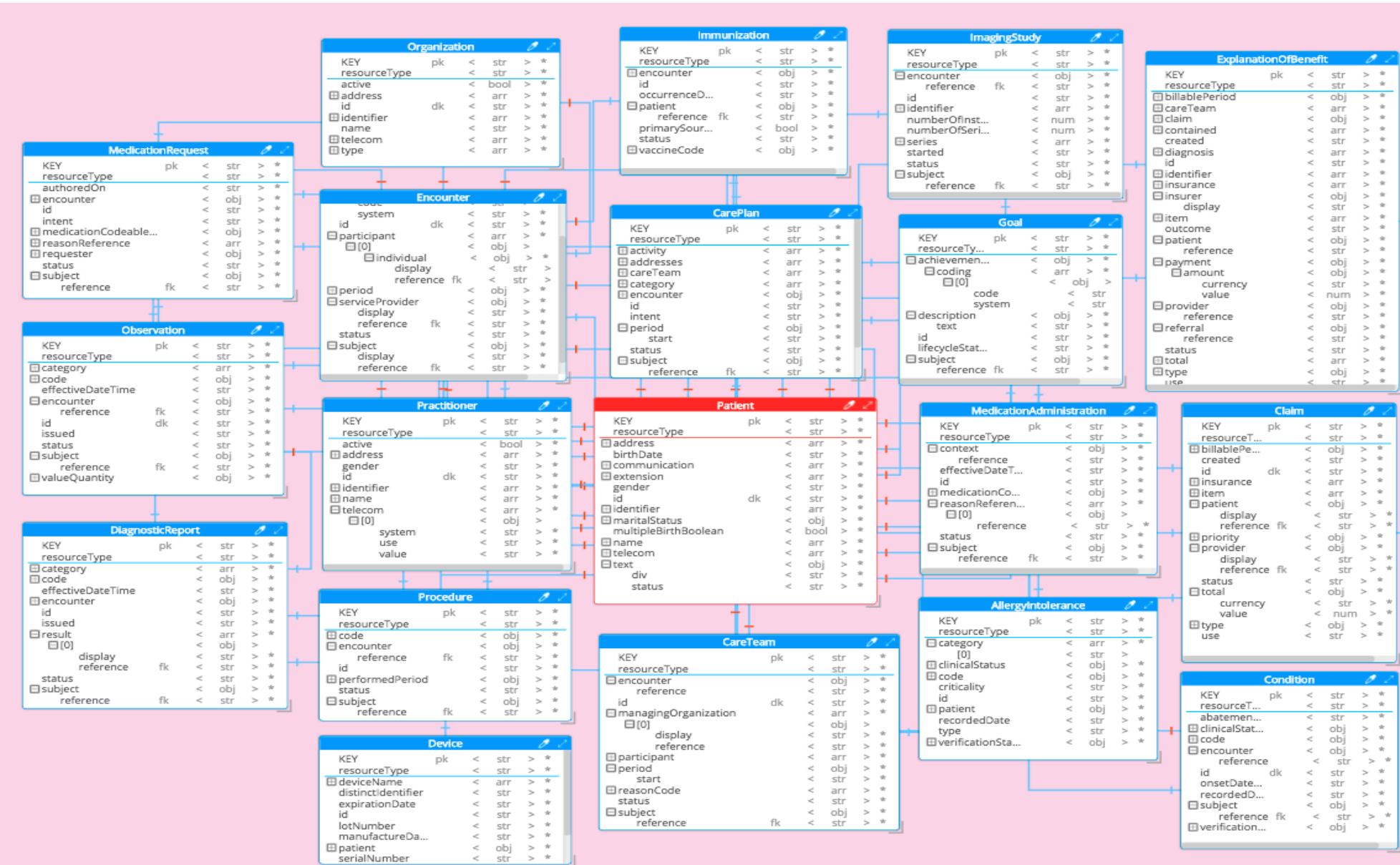


Data Models

Data models are also known as relational models/schemas (in DB theory and ER modelling).

Subset of the HL7 FHIR data model for clinical data exchange.

Source:
<https://www.couchbase.com/blog/fhir-data-model-with-couchbase-n1ql/>



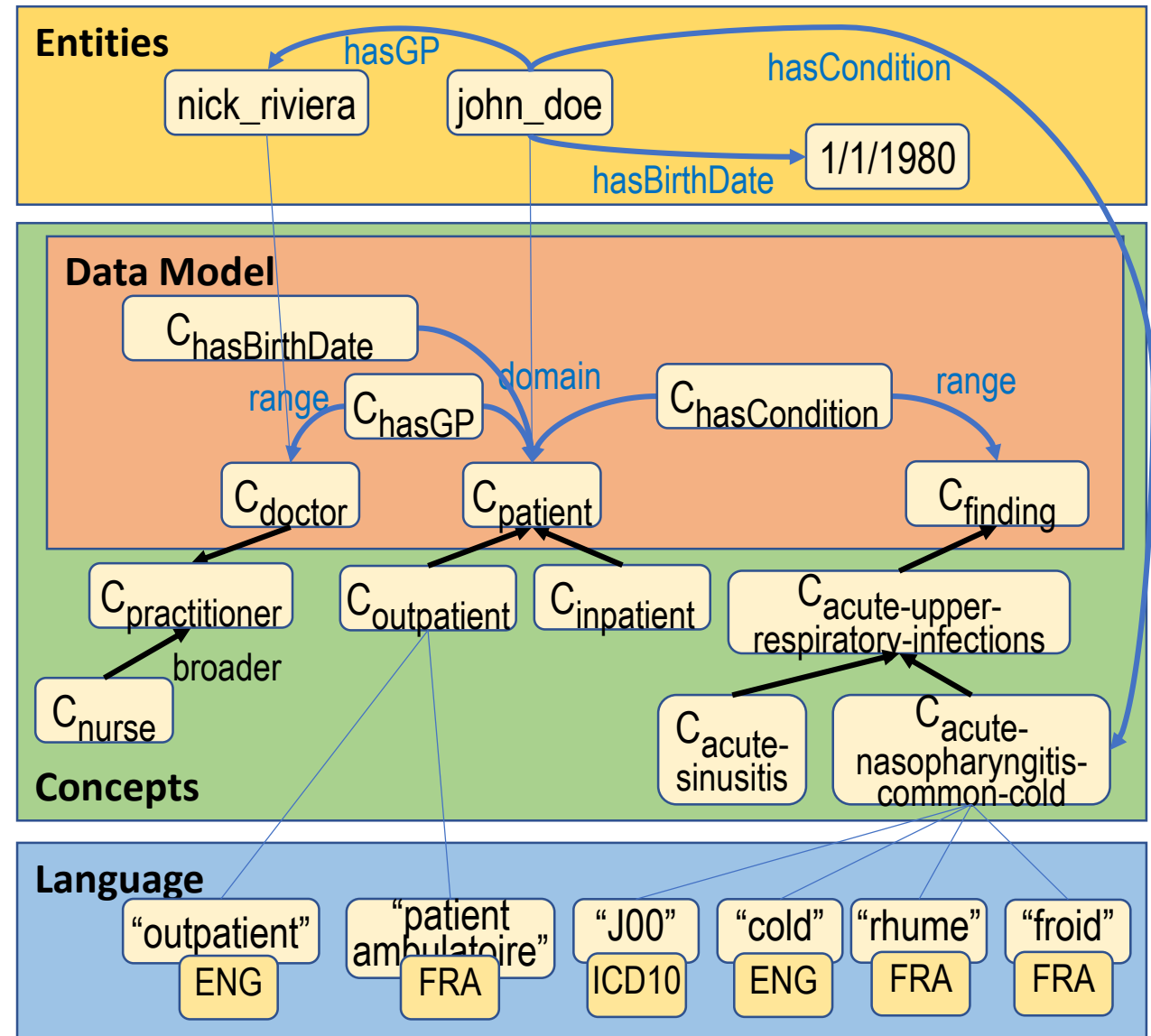
Layers of Knowledge: Common KO Schemes

Entities or individuals describe data instances of interest via attributes and relations.

Data models or schemas designate a subset of concepts that describe entities of primary interest, and provide a predefined way of describing them via attributes and relationships to other entities. Data models and concept hierarchies are often designed separately.

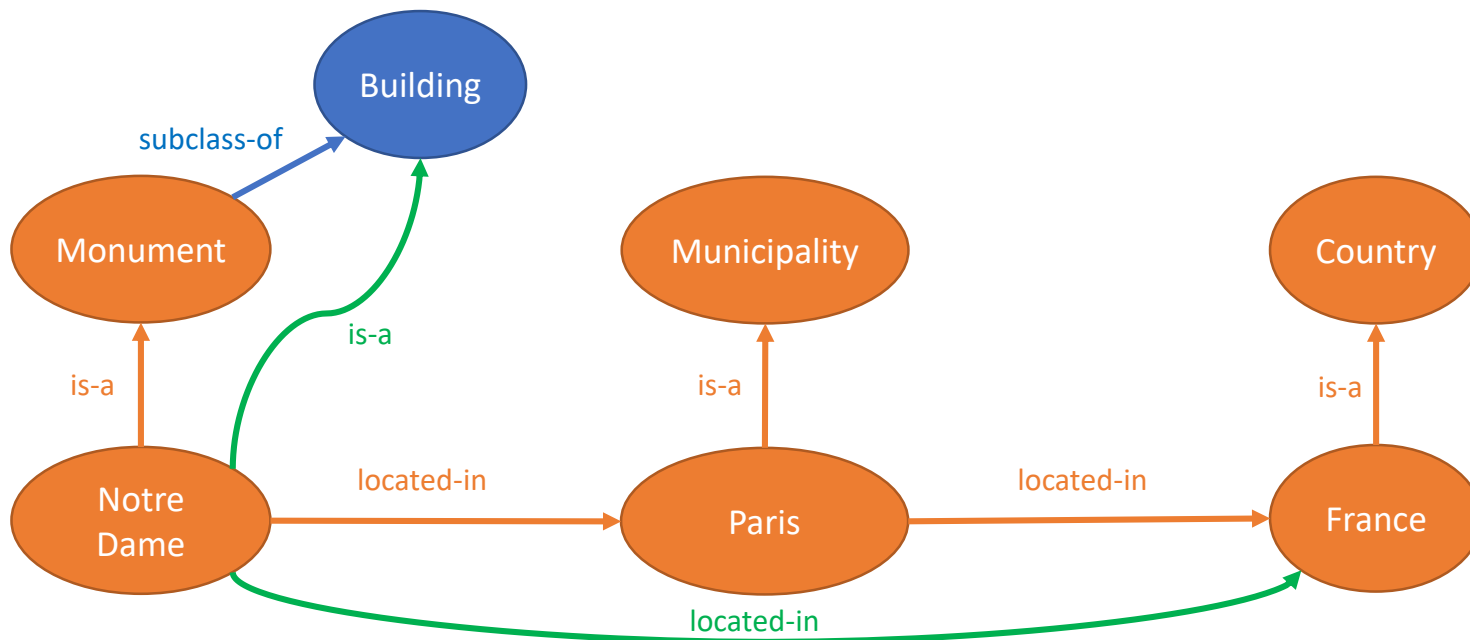
Concepts represent ideas independently of language. (Here, we are using English for understandability.) A complex domain, such as healthcare, can easily define hundreds of thousands of concepts. Concepts are related to each other through (hierarchical or associative) relationships.

Human knowledge has been primarily described verbally => **language** remains the fundament of KR. A domain can have hundreds of thousands of specialised terms per language.



Knowledge Graphs (KG)

- A major subfield within symbolic AI research, since the beginning (semantic networks).
- Formalised in diverse ways: frames, ontologies, concept nets, etc.
- Data graph = data structured as a graph.
- Knowledge graph = data graph + background knowledge, from which new, unseen data can be inferred.
- Inference is implemented as algorithms over the graph.



Stated facts:

- Notre Dame is a Monument
- Notre Dame is located in Paris
- Paris is a Municipality
- France is a Country
- Paris is located in France

Axioms for inference:

- every Monument is a Building
- A located-in B, B located-in C => A located-in C

Inferred facts:

- Notre Dame is a Building
- Notre Dame is located in France

How do Knowledge Graphs Help with Managing Heterogeneity?

- A KG (as we will define them) is a **formal, less ambiguous** representation of data.
- It **makes explicit** many of the implicit assumptions underlying the data (e.g. “Paris” means “Paris, France” and not “Paris, Texas”).
- It is **interpretable** both by humans and machines that can **reason** with it automatically.
- It is **extensible** by new data and knowledge without any need for refactoring.
- **Existing human knowledge** can be expressed with KGs in a straightforward way, and many such resources exist and can be reused:

GeoNames

Geographic entities and their relationships



Data model for healthcare interoperability



Facts extracted from Wikipedia



Community-sourced formal facts



Unified Foundational
Ontology

Fundamental concepts describing the world

CIDOC
CRM

CONCEPTUAL
REFERENCE
MODEL



Terminology and data model for cultural heritage



Medical terminology

Knowledge Graphs in Industry

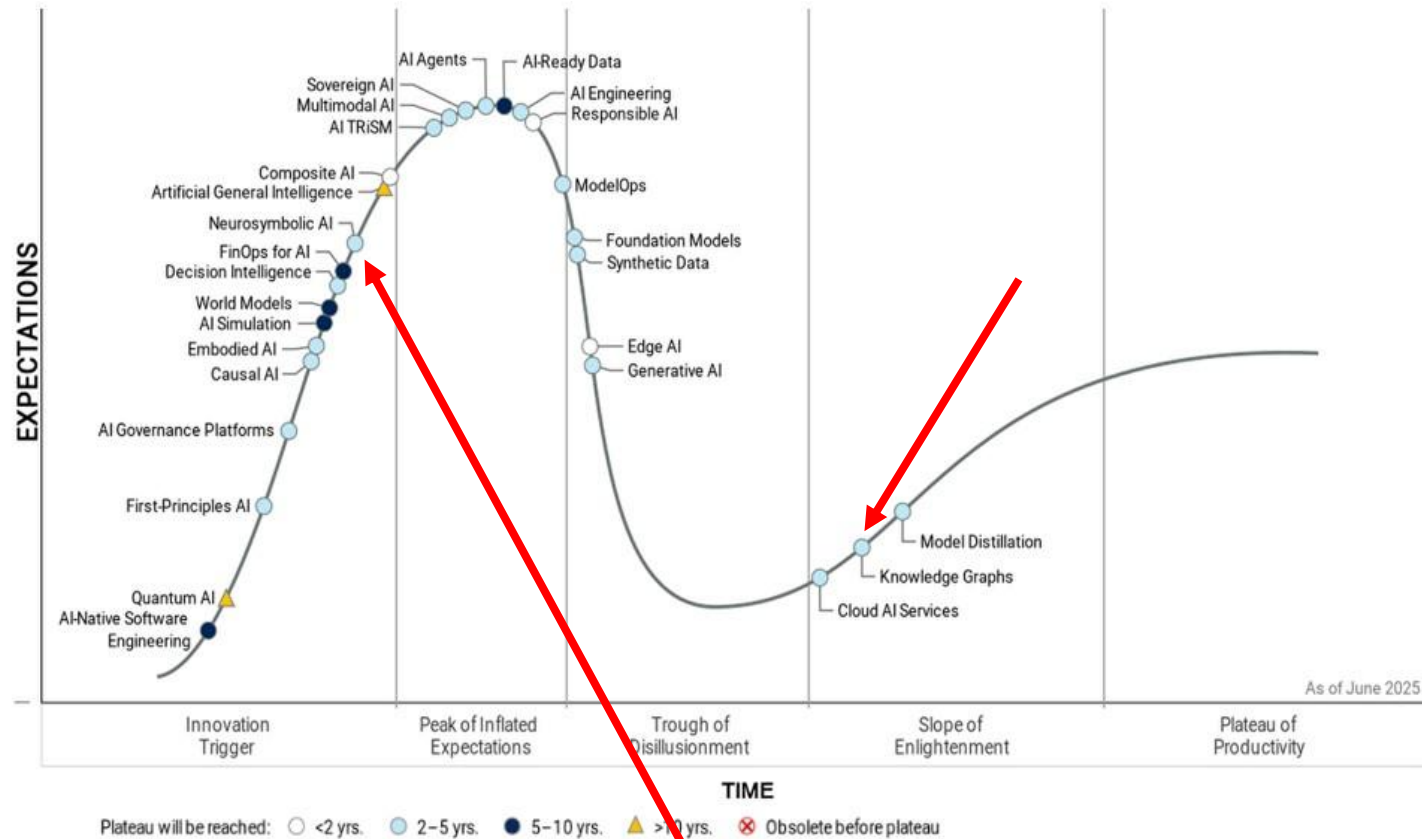


	DATA MODEL	SIZE OF THE GRAPH	DEVELOPMENT STAGE
Microsoft	The types of entities, relations, and attributes in the graph are defined in an ontology.	~2 billion primary entities, ~55 billion facts	Actively used in products
Google	Strongly typed entities, relations with domain and range inference	1 billion entities, 70 billion assertions	Actively used in products
Facebook	All of the attributes and relations are structured and strongly typed, and optionally indexed to enable efficient retrieval, search, and traversal.	~50 million primary entities, ~500 million assertions	Actively used in products
eBay	Entities and relation, well-structured and strongly typed	Expect around 100 million products, >1 billion triples	Early stages of development and deployment
IBM	Entities and relations with evidence information associated with them.	Various sizes. Proven on scales documents >100 million, relationships >5 billion, entities >100 million	Actively used in products and by clients

Noy et al.: Industry-scale Knowledge Graphs: Lessons and Challenges: Five diverse technology companies show how it's done. Communications of the ACM, 2019.
Amit Singhal (Google): Introducing the Knowledge Graph: things, not strings. <https://blog.google/products/search/introducing-knowledge-graph-things-not-strings/>
Amazon knowledge graph tutorial: https://naixlee.github.io/Product_Knowledge_Graph_Tutorial_KDD2021/

Knowledge Graphs on the Gartner Hype Cycle

Gartner Hype Cycle for Artificial Intelligence, 2025



- KGs are a mature technology that survived hype and disillusionment.
- Their strengths and limits are now understood and are used accordingly.
- One of the hottest topics in current research: combining KGs and deep learning.

Data and Knowledge Graph: Formal Definition

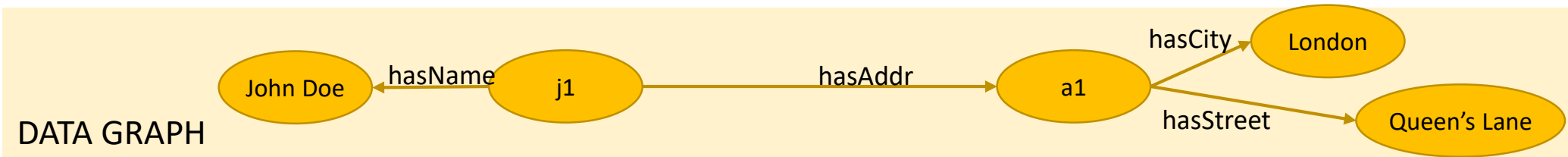
Highly recommended reading material on knowledge graphs (the entire book is available for free):

Hogan et al.: Knowledge Graphs. Morgan & Claypool Publishers, 2021. <https://kgbook.org/>

- Knowledge = data + background knowledge, from which new, unseen data can be inferred.
- When our data is structured as a graph, we call it a **data graph**.

Definition: a **data graph** is a directed, node-labelled and edge-labelled graph $G = (V, R, E)$ where

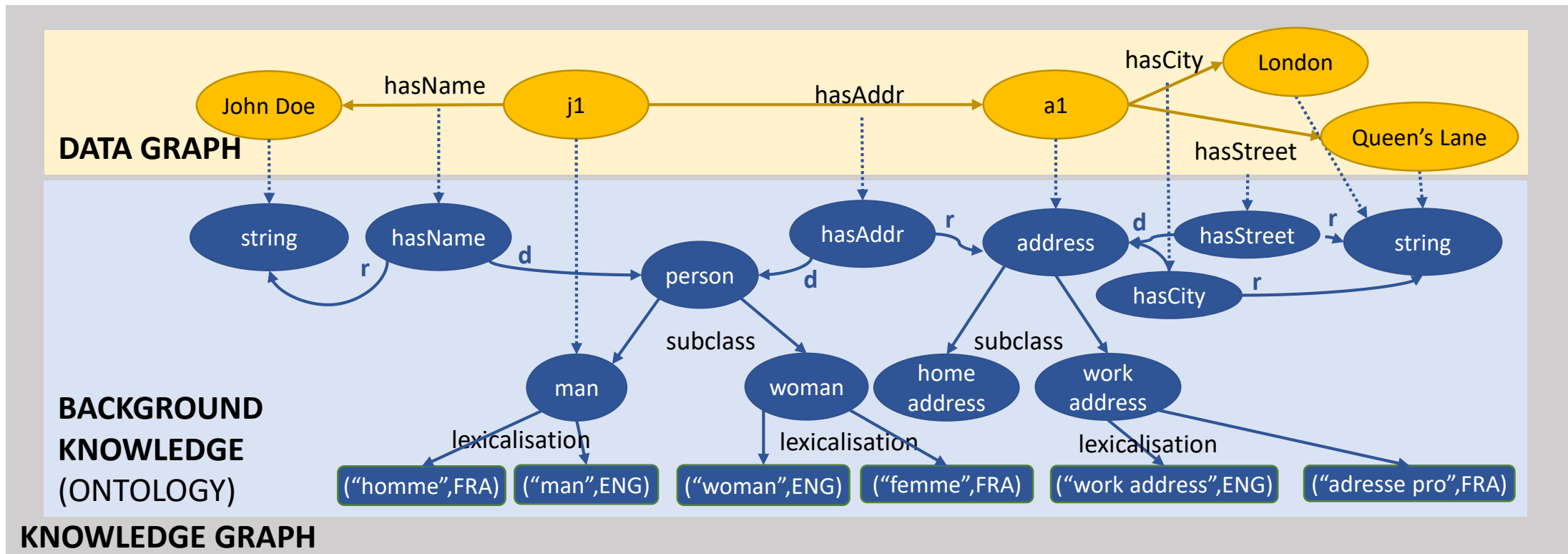
- $V = \{\text{string}\}$ are vertices (nodes);
- $R = \{\text{string}\}$ are edge (relation) labels;
- $E \subseteq V \times R \times V$ are labelled edges.



- Our definition is compatible with RDF specifications.

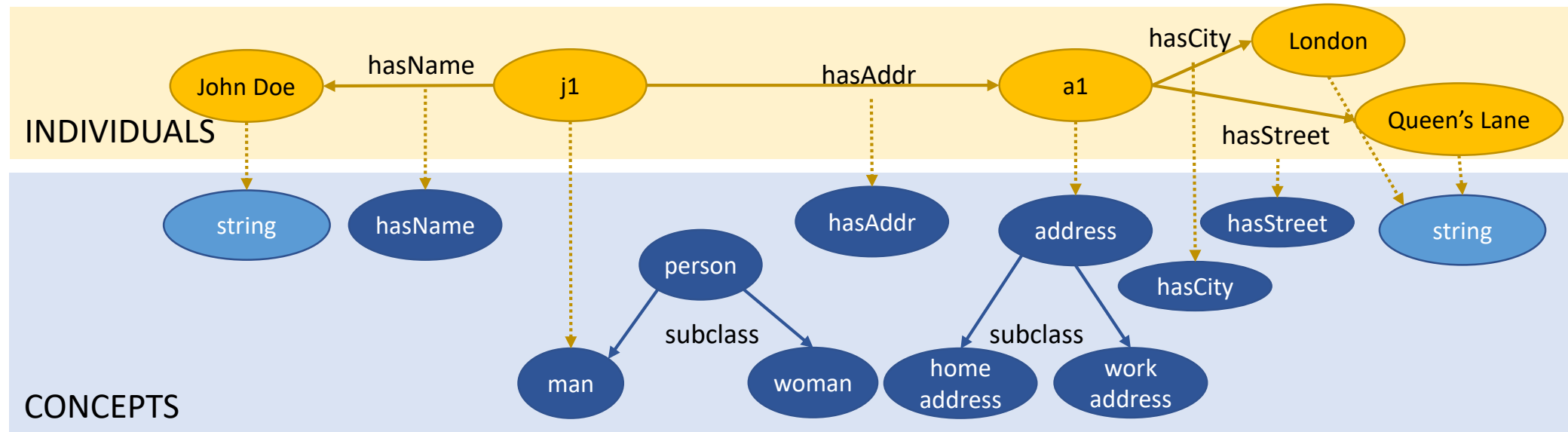
KG: Towards a Formal Definition

- The terms “knowledge graph” and “ontology” are not always used in a precise way, especially in the industry. We propose the following convention.
- Knowledge = data + background knowledge, from which new, unseen data can be inferred.
- A **knowledge graph** (also called **ontology** in the context of the Semantic Web) is a data graph that is extended with background knowledge for purposes of inference.
- Background knowledge is general and reusable across datasets => it is split from the data graph.



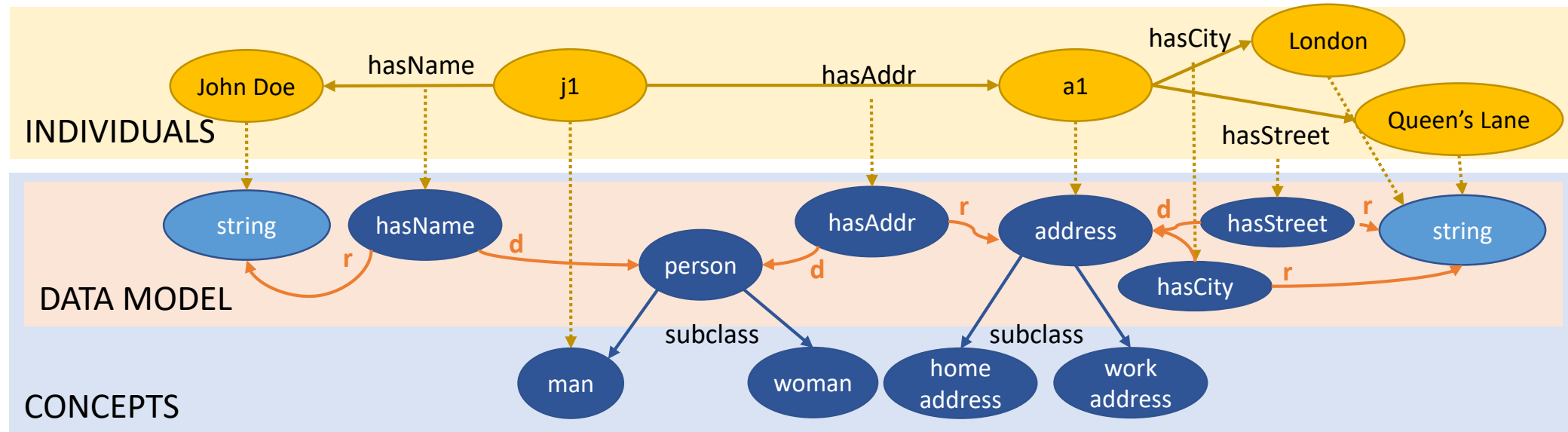
KG: Concept Layer

- As in description logic, the set of nodes V is split into **concepts** C and **individuals** I : $V = C \cup I$;
- individuals and their relations are **typed** by concepts: for every $i \in I$ or $r \in R$ there is an edge $(i, \text{"type"}, c)$ or $(r, \text{"type"}, c)$ meaning that the type of i or r is c (dotted arrows below);
- **literals** are individuals that are **typed by datatype concepts** such as *string*, *integer*, *date*, etc.;
- concepts form a network, e.g. one can build concept hierarchies via the **subclass** relation;
- everything stated or inferred about a concept c automatically applies to all individuals typed by it.



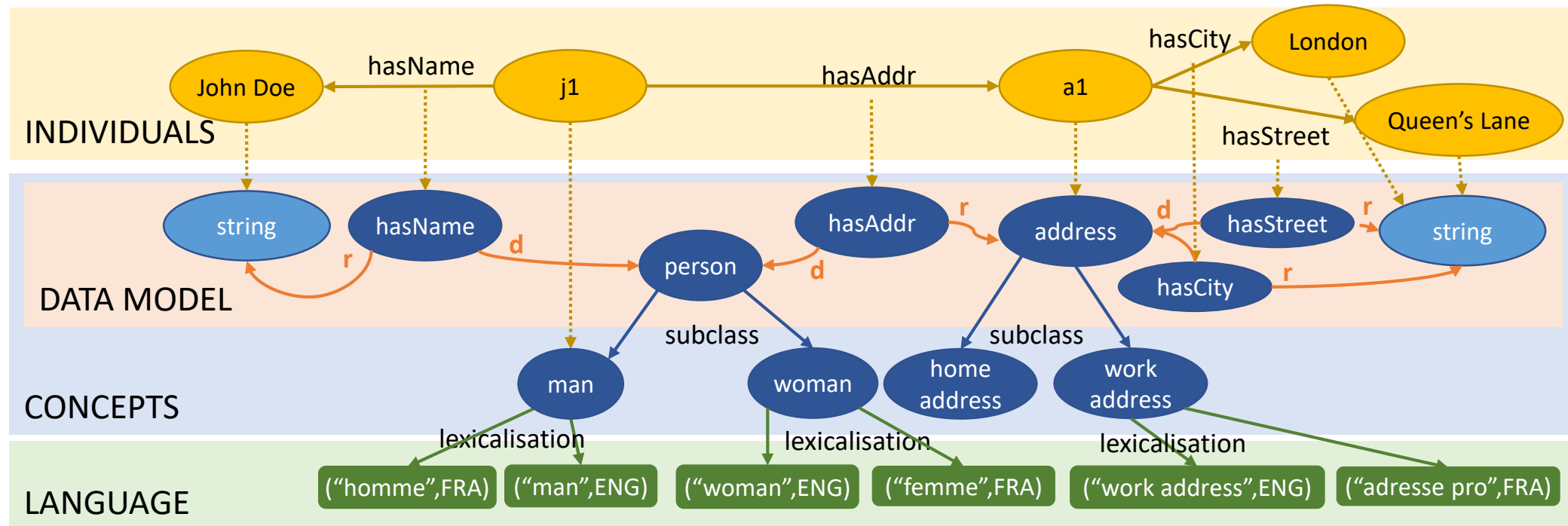
KG: Data Model Layer

- The **data model** D is a small subset of the concept layer.
- **The data model consists of the most general and most frequently instantiated types.**
- A concept that represents a relation type has a **domain (d)** and a **range (r)** relation to the types of its source and target nodes.
- Relations of which the range is a datatype are also called **attributes** (below, hasName, hasCity, and hasStreet are attributes).



KG: Language Layer

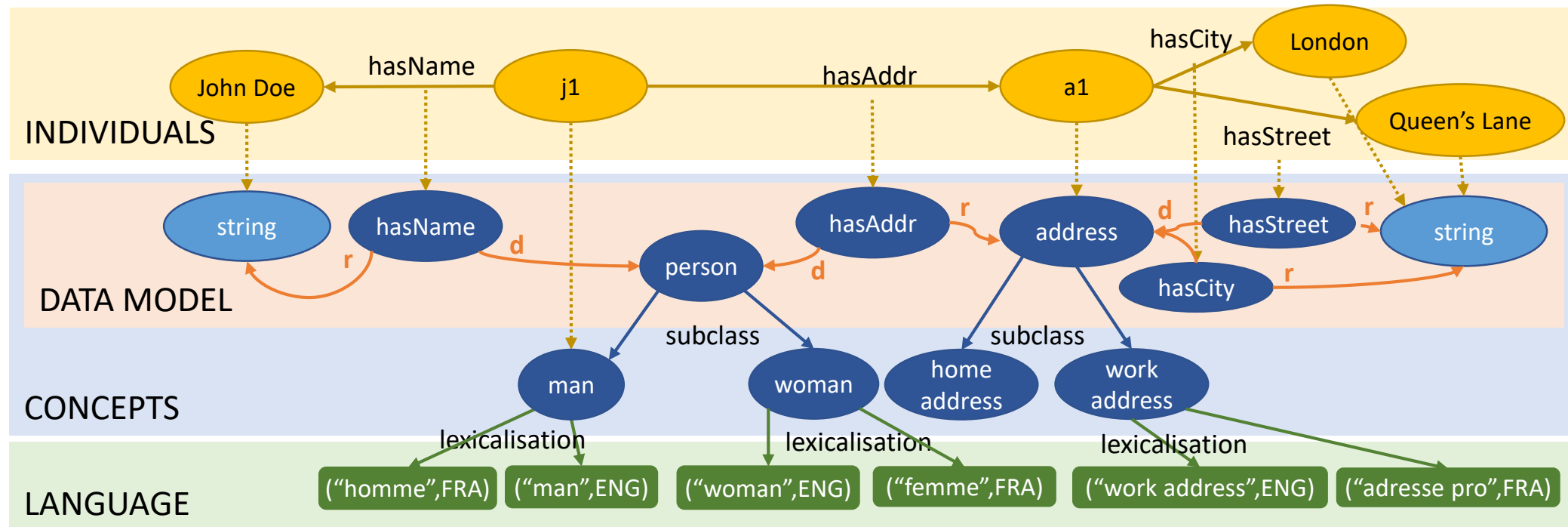
- The **language layer** L , in its simplest form, is a set of lexicalisation relations between concepts and lexemes: $L = \{(c, l)\}$ where:
 - c is a concept;
 - l is a lexeme, where $l = (\ell, \tau)$:
 - ℓ is a string (a word, term, or expression),
 - τ is a language tag (e.g. two-letter ISO 639-1 or three-letter ISO 639-3 tags).



Knowledge Graph: Formal Definition

In our definition, a knowledge graph KG is a graph consisting of an individual layer I , a data model layer D , a concept layer C , and a language layer L :

$$KG = \langle I, D, C, L \rangle.$$



Knowledge Graphs in the Semantic Web

The Semantic Web languages RDF, RDFS, and OWL can represent knowledge graphs as defined before.

KG layer	KG element	Example	Implementation in RDFS/OWL
data graph	individual	a specific patient at a hospital	chu:Patient/123456
	literal	"John Doe"	"John Doe"^^xsd:string
	relation	the patient's name is "John Doe"	chu:Patient/123456 chu:name "John Doe"^^xsd:string
concept	concept	Patient is a concept	fhir:Patient rdf:type owl:Class
	type of an individual	John Doe is a patient	chu:Patient/123456 rdf:type fhir:Patient
data model	attribute	patients have a name	hasName rdf:type owl:DatatypeProperty hasName rdfs:domain fhir:Patient hasName rdfs:range xsd:string
	relation	patients are treated by practitioners	treatedBy rdf:type owl:ObjectProperty treatedBy rdfs:domain fhir:Patient treatedBy rdfs:range fhir:Practitioner
language	lexicalisation	in Italian, patient is "paziente"	chu:Patient/123456 rdfs:label "paziente"@it

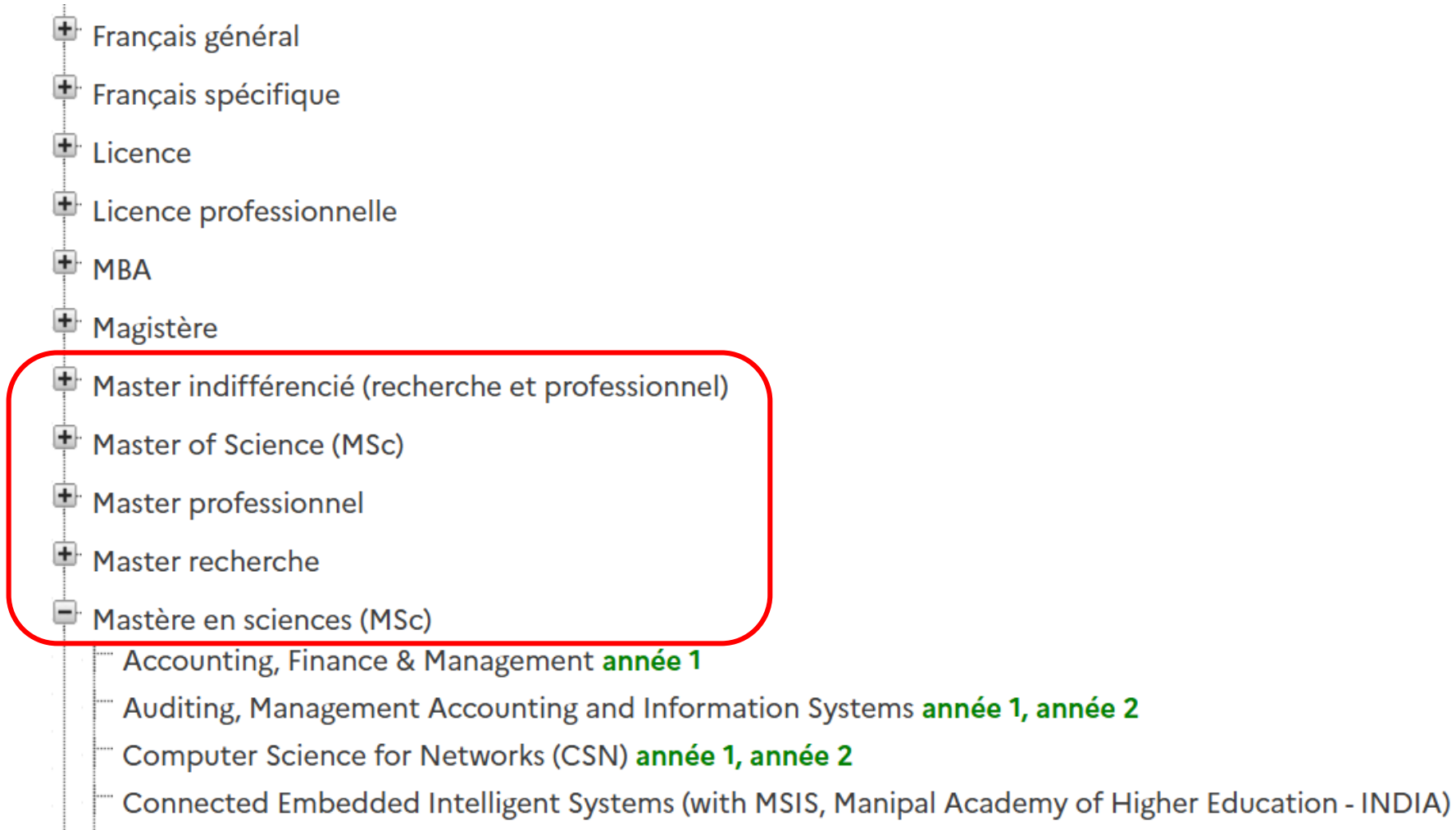
Knowledge Processing Heterogeneity in Knowledge

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Heterogeneity in Knowledge

- Background knowledge helps us tackle heterogeneity in data.
- Bad news: **what if knowledge itself is heterogeneous?**
- Of course it is heterogeneous, as any non-trivial system designed by humans.
- Some of the main reasons:
 - **implicit assumptions** about the meaning of knowledge that not all users share;
 - **differing intentions and viewpoints** about the world lead to **ad-hoc or incomplete knowledge**;
 - **explicit disagreement** over how to formalise a domain;
 - **parallel development** of formalisms (e.g. national standards);
 - **evolution** of human knowledge, needs, viewpoints leads to diverging formalisations.
- Why is it a problem? Globalisation and the Internet => incentives to interoperate.

Example: catalogue Études en France, 2025



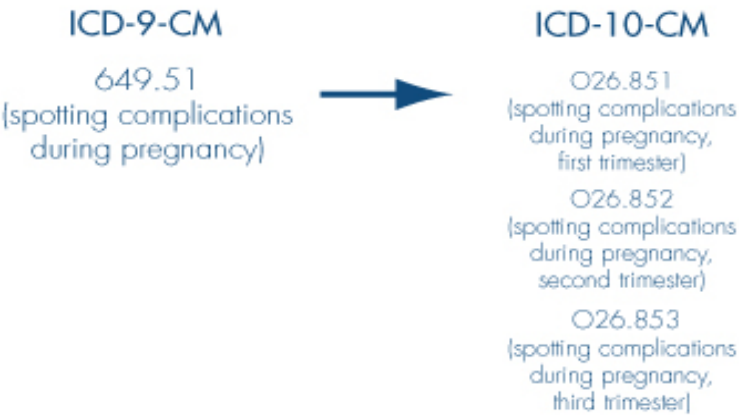
- A large catalogue with thousands of training programmes.
- Heterogeneity within the classification of training types is a major problem both for schools (visibility) and for candidates (finding trainings).

Example: Evolution of Classification (Heterogeneity in Time)

- Incremental evolution of knowledge representation (e.g. North American Industry Classification System, 2017 vs 2022);
- major evolution of a classification system (e.g. version change from ICD-9 to ICD-10);
- evolution of knowledge itself (e.g. COVID-19 introduced into the International Classification of Diseases v10 under emergency codes U07.1 and U07.2.)



ICD-9	ICD-10
3-5 characters in length	3-7 characters in length
Approximately 13,000 codes	Approximately 68,000 available codes
First digit may be alpha (E or V) or numeric; digits 2-5 are numeric	Digit 1 is alpha; digits 2 and 3 are numeric; digits 4-7 are alpha or numeric (alpha digits are not case sensitive)
Limited space for adding new codes	Flexible for adding new codes
Lacks detail	Very specific
Lacks laterality	Has laterality (i.e., codes identifying right vs. left side of the body)
Use same code for every visit	Has possibility of identifying initial encounter, subsequent encounter, or sequela
Only 4 codes were reported on a claim form	Up to 12 codes can be reported on a claim form



Example: National Standards (Heterogeneity in Space)

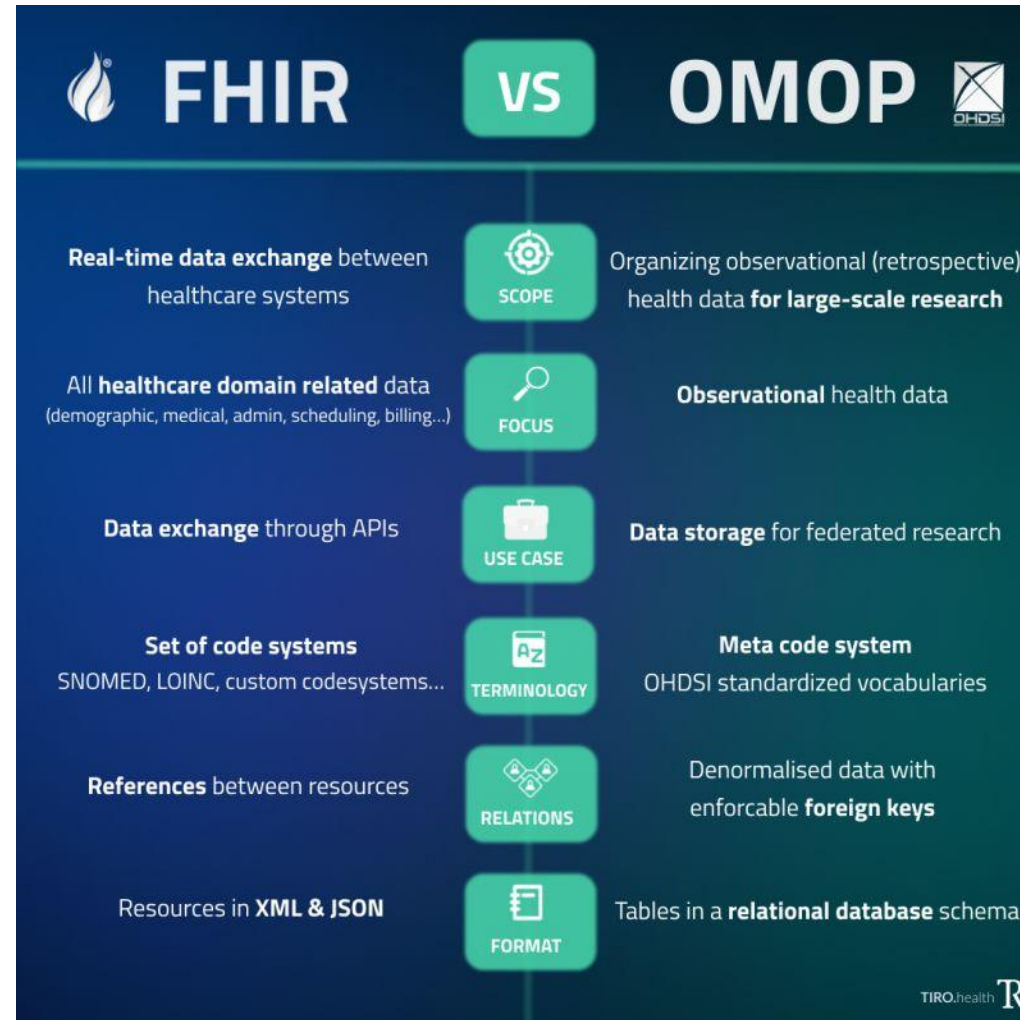
- North American Industry Classification System:
 - NAICS Canada;
 - NAICS USA;
 - SCIAN Mexico.
- Mostly aligned, but not fully, e.g. under Public Administration:

NAICS USA	SCIAN Mexico	NAICS Canada
Executive, Legislative, and Other General Government Support	Actividades legislativas, gubernamentales y de impartición de justicia	Federal government public administration
Justice, Public Order, and Safety Activities		Provincial and territorial public administration
Administration of Human Resource Programs		Local, municipal and regional public administration
Administration of Environmental Quality Programs		Indigenous public administration
Administration of Housing Programs, Urban Planning, and Community Development	Organismos internacionales y extraterritoriales	International and other extra-territorial public administration
Administration of Economic Programs		
Space Research and Technology		
National Security and International Affairs		

Example: Differing Viewpoints (Heterogeneity in Perspective)

FHIR: for primary use of health data (care),
in a context of interoperability

OMOP: for secondary use of health data (research)



Knowledge Alignment: Intuition



- Two knowledge graphs may represent truth in two different ways.
- Knowledge alignment creates a “bridge” between the two graphs so that data is able to “cross” from one to the other.
- The end result is a single big knowledge graph that includes both previous ones, plus the bridge.

Knowledge Alignment: Definition

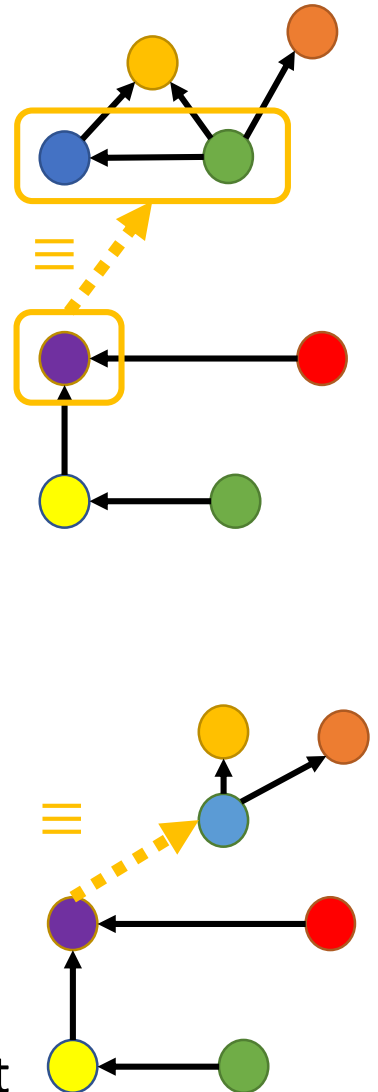
- **Alignment:** given two knowledge graphs KG_A and KG_B , and two layers L_A and L_B of the same kind (individual, concept, data model, language) from KG_A and KG_B , an **alignment** $\mathcal{A}_{AB}$ is a set of qualified **mappings** (a, G_A, G_B) between subgraphs $G_A \subseteq L_A$ and $G_B \subseteq L_B$:

$$\mathcal{A}_{AB} = \{(a, G_A, G_B)\}$$

- The qualifier a is a logical axiom over the elements in G_A and G_B .
- **Knowledge alignment** is the (automated or human-driven) operation that, given A and B , provides the alignment $\mathcal{A}_{AB}$ and subsequently materialises the knowledge graph KG_{AB} where

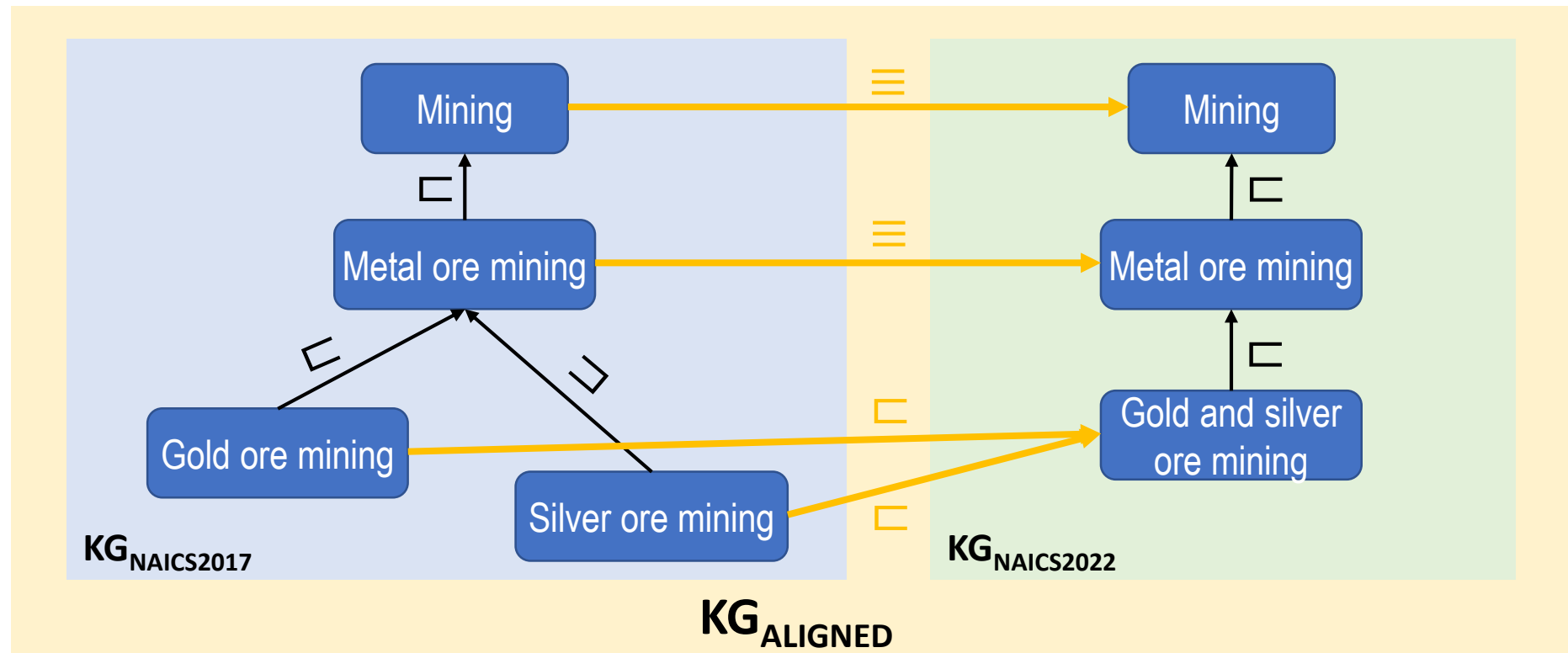
$$KG_{AB} = KG_A \cup \mathcal{A}_{AB} \cup KG_B$$

- $\mathcal{A}_{AB}$ is the “bridge” between KG_A and KG_B
- A mapping is **atomic** if $|G_A| = 1$ and $|G_B| = 1$ (between single nodes or between edges).
- In practice, if a mapping is not atomic ($|G_A| > 1$ or $|G_B| > 1$) then we decompose it into a set of atomic mappings. This often requires the **transformation** of G_A or G_B .



Example: Formalisation & Alignment of Taxonomies

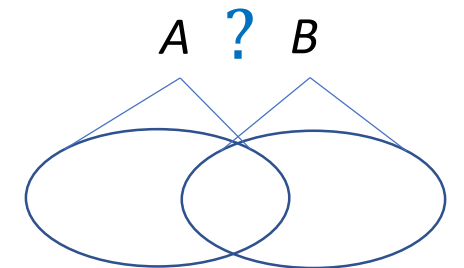
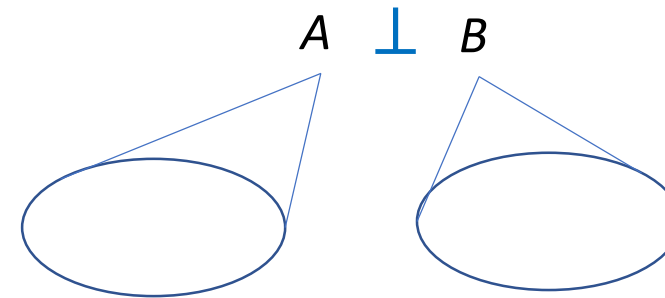
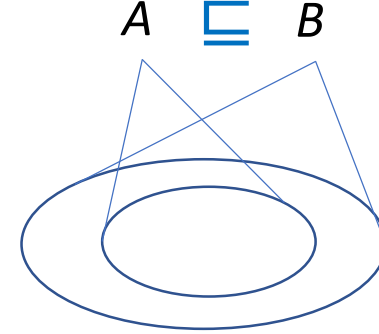
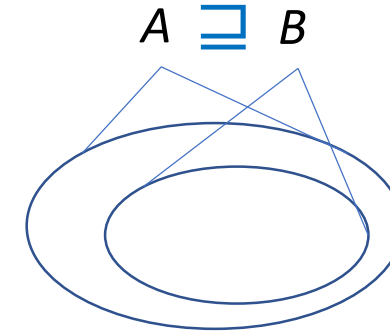
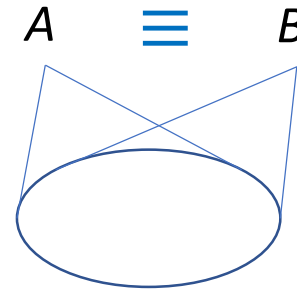
Let's become concrete! You have half of your dataset of businesses classified under NAICS 2017, the other half under NAICS 2022. What do you do?



- The final knowledge graph contains both versions, aligned: you can now query your dataset using either version of NAICS, the underlying reasoner will interpret the mapping axioms under the hood.
- E.g. a company classified under “Gold ore mining” in NAICS 2017 will also be classified under “Gold and silver ore mining” in the aligned KG.

The Semantics of Mappings: Extensional Analysis

- What kinds of logical axioms can we use for mapping?
- From a set-theoretical perspective, between two classes A and B , five kinds of alignments are possible, based on their extensions (i.e. the individuals they classify):
 - $\equiv$ **equivalence**: they classify the same individuals;
 - $\perp$ **disjointness**: they have no common individuals;
 - $\supseteq \sqsubseteq$ **containment**: one is a broader set than the other (in either sense);
 - $?$ **overlap**: the most general scenario, not describable by a single logical operator.



The Semantics of Mappings: Examples

≡ equivalence

In a dataset, you can safely replace the source by the target, or vice versa.

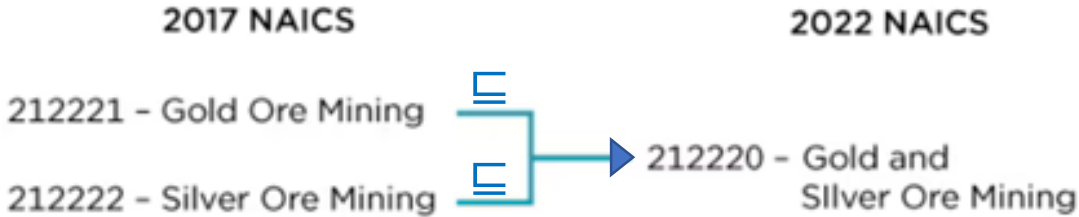
SCIAN Mexico	NAICS Canada
Actividades legislativas, gubernamentales, de impartición de justicia y de organismos internacionales y extraterritoriales	Federal Administration
Organismos internacionales y extraterritoriales	International and other extra-territorial public administration

⊥ disjointness

In a dataset, classifying an instance under both categories leads to a logical contradiction.

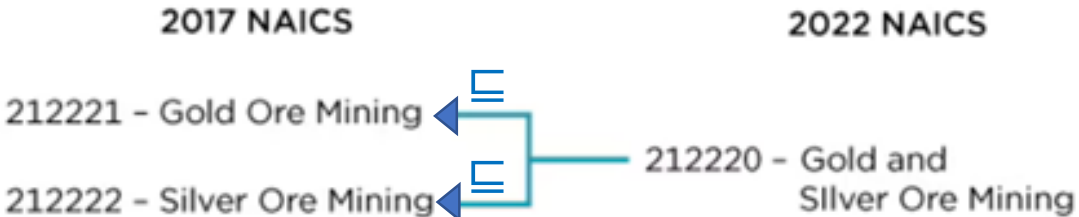
Classification 1	Classification 2
Mining	Extraction
Gold Mining	Extraction of metals except for gold

⊆ narrower than



In a dataset, you can safely replace the source by the target, but you potentially lose information.

⊇ broader than



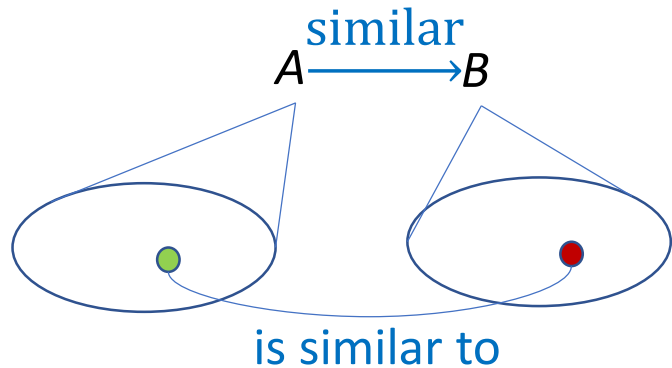
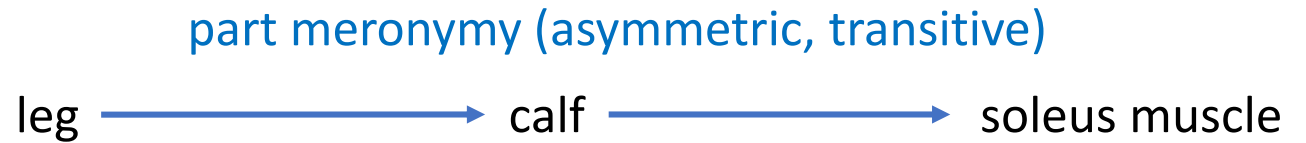
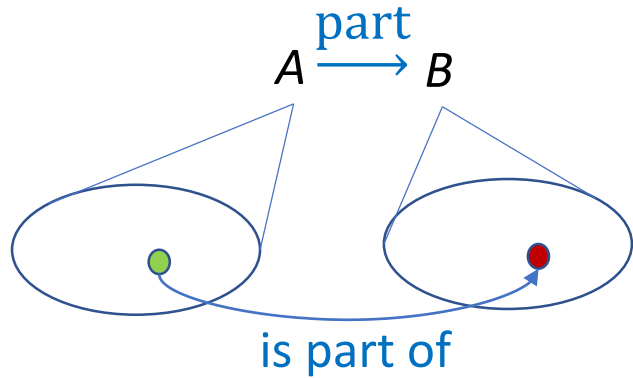
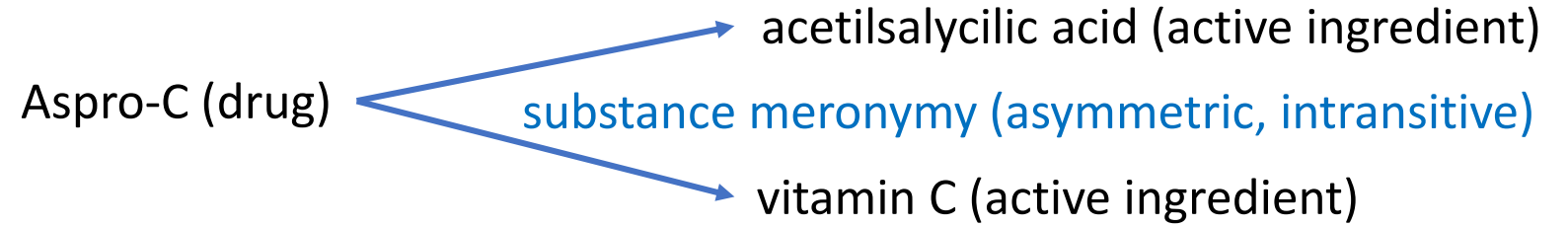
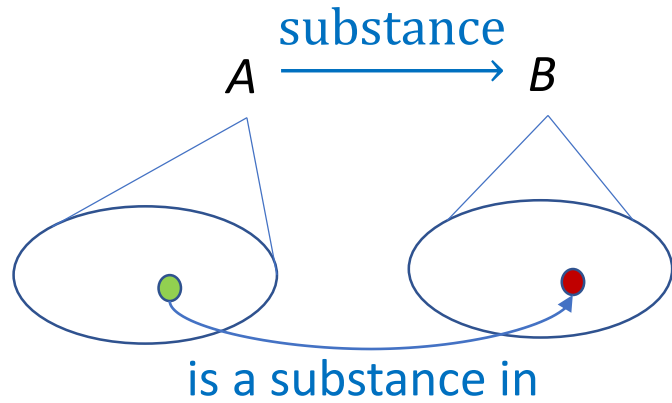
In a dataset, you cannot mechanically replace the source by the target, as you may introduce incorrect information. More evidence needs to be gathered from the context (here: gold or silver).

Mappings in the Semantic Web

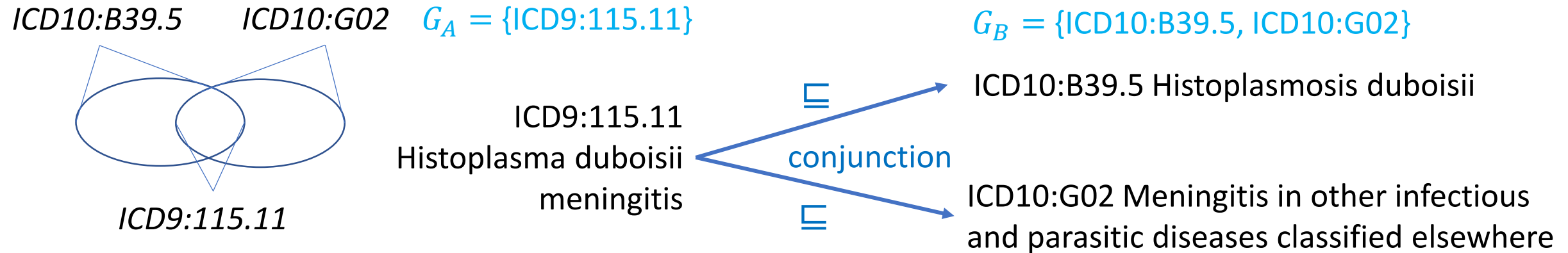
- The Semantic Web languages offer standard relations to express semantic mappings.

Mapping operator	Between concepts	Between individuals
≡ Equivalence	owl:equivalentClass	owl:sameAs skos:exactMatch
⊇ Broader/narrower	rdfs:subClassOf	skos:broadMatch skos:narrowMatch
⊥ Disjointness	owl:disjointWith	owl:differentFrom

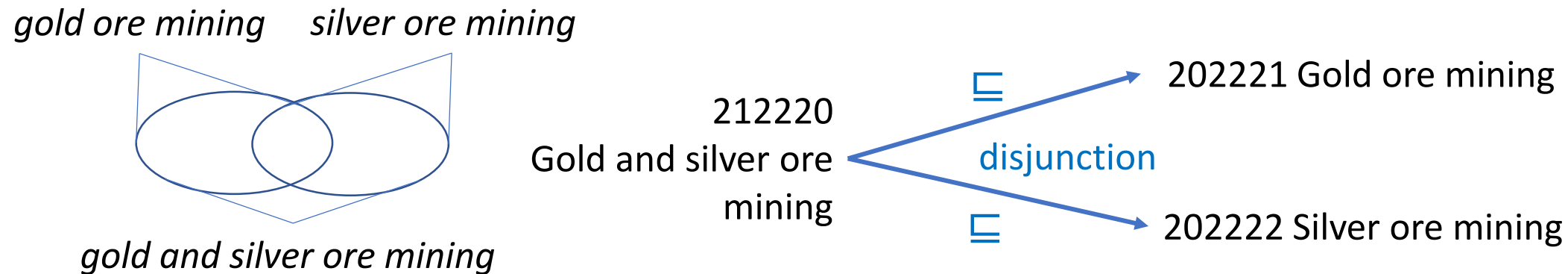
The Semantics of Mappings: Other Mapping Operators



The Semantics of Mappings: Complex Axioms



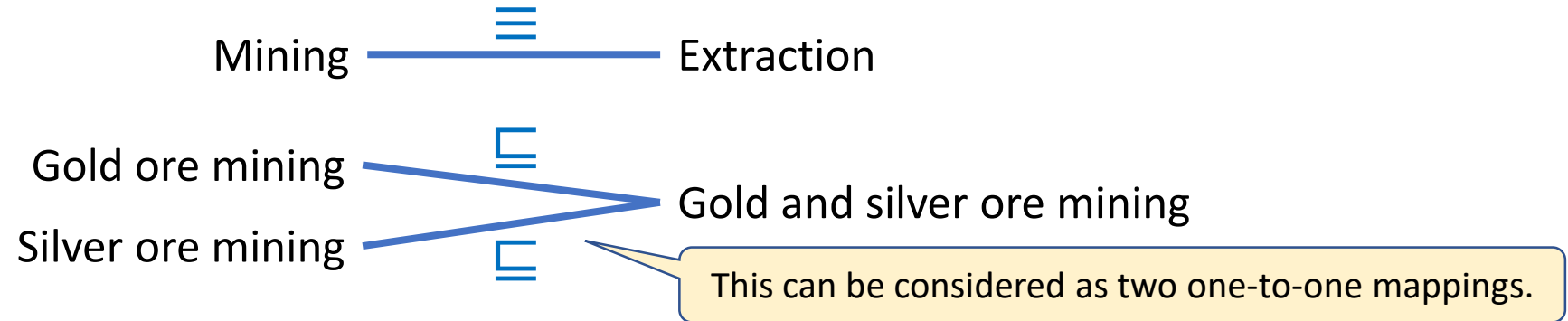
$$a = (ICD9:115.11 \equiv ICD10:B39.5 \sqcap ICD10:G02)$$



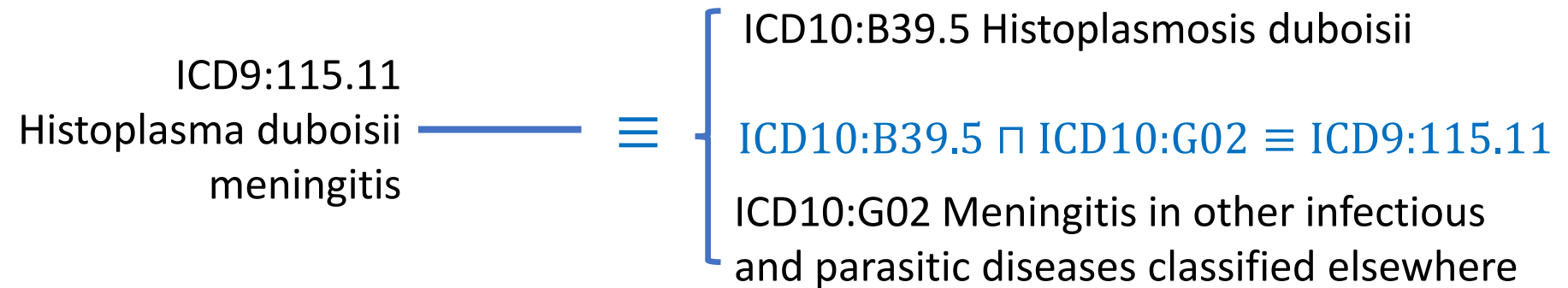
$$a = (202220 \equiv 202221 \sqcup 202222)$$

The Semantics of Alignment: Cardinality

one-to-one
 $|G_A| = |G_B| = 1$



one-to-many
 $|G_A| = 1, |G_B| > 1$



many-to-many
 $|G_A| > 1, |G_B| > 1$

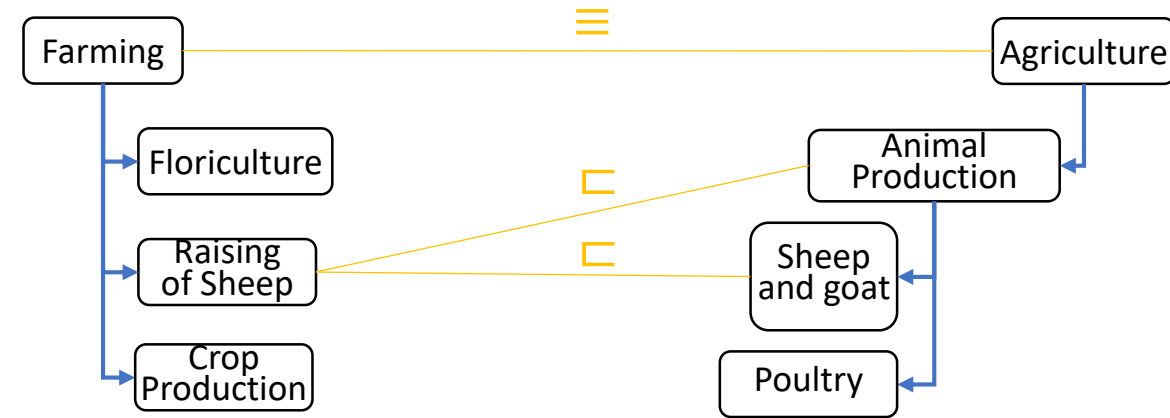


one-to-zero
 $|G_A| = 1, |G_B| = 0$



Automated Ontology Alignment

- Goal: find correspondences between the elements of two input ontologies.
- “Ontology” is understood in a broad sense: sometimes, natural-language taxonomies and even terminologies are included.
- A 20-year-old, mature field of research (see <http://www.ontologymatching.org/>):
 - until the mid-2000s:
 - string-similarity-based methods (surface comparison of NL labels),
 - cannot deal with linguistic heterogeneity, e.g. synonyms, cannot align across languages, etc.
 - until the mid-2010s:
 - word-meaning-based methods,
 - graph-based (structural) methods,
 - instance-based (extensional) methods;
 - since the mid-2010s:
 - cross-lingual methods,
 - word-vector-based methods,
 - language-model-based methods.



The Semantics of Mappings: Similarity Score

animal husbandry	$a = 0.9$	raising of cattle
animal husbandry	$a = 0.55$	farming
animal husbandry	$a = 0.23$	growing of cereals

- Automated alignment methods often return similarity scores as opposed to logical axioms, e.g. between 0 and 1.
- If similarity > an empirically fixed threshold (e.g. 0.9), then a predefined axiom is stated (e.g. equivalence).
- However, similarity scores are not semantic: such mappings need to be verified and interpreted by a human expert.
- In an RDF-based knowledge graph, less-than-one similarity mappings can be materialised, for example, via `skos:closeMatch` or `skos:relatedMatch` properties.

Ontology Alignment Techniques

Technique	Works well on	Does not work on
String-based, e.g. fuzzy matching (until the 2000s)	minor grammatical differences (singular vs plural, word order change, etc.) <i>Crop farming</i> $\approx$ <i>Farming of crops</i> e.g. Levenshtein, Jaccard similarity	<i>Farming</i> = <i>Agriculture</i> because the strings are too different <i>Farming</i> $\neq$ <i>Charming</i> because the strings are too similar
Knowledge-based, e.g. semantic matching	<i>Farming</i> $\equiv$ <i>Agriculture</i> because they are synonyms. <i>Raising of sheep</i> $\sqsubset$ <i>Animal production</i> because the following formula is satisfiable: <i>Raising</i> $\wedge$ <i>Sheep</i> $\rightarrow$ <i>Animal</i> $\wedge$ <i>Production</i> considering the background knowledge: <i>Raising</i> $\rightarrow$ <i>Production</i> $\wedge$ <i>Sheep</i> $\rightarrow$ <i>Animal</i> $\wedge$	<i>Astronomy</i> $\equiv$ <i>Celestial mechanics</i> <i>Citrus groves</i> $\equiv$ <i>Growing of citrus fruits</i> because they intend to mean the same without expressing the same concept
Neural networks, e.g. word embeddings, LLMs	When approximate matching is adequate, based on underlying meaning	Large-scale data, high-precision contexts

- Automated ontology alignment is supposed to reduce workload on humans but, due to the complexity of semantic alignment, human oversight is unavoidable.
- => Knowledge alignment is a labour-intensive task.

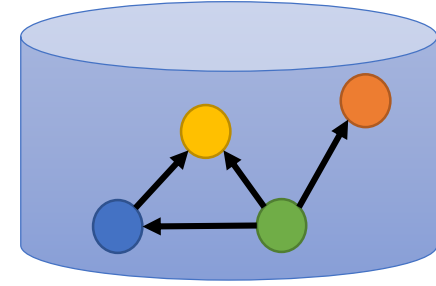
Knowledge Processing

Knowledge Graph Exploitation

Gábor Bella, Yannis Haralambous

Direct KG Exploitation

- How to query or search over a graph?
- Query languages:
 - SPARQL:
 - W3C / Semantic Web standard;
 - matches and returns RDF triples;
 - can return results inferred by an underlying reasoner.
 - Other contenders:
 - Cypher: proprietary language of the Neo4J graph DB;
 - GQL (Graph Query Language): recent standardisation effort, related to Cypher, simpler to use than SPARQL.
- Information retrieval:
 - simple text-based search: the graph DB indexes all strings within the graph:
 - IRIs;
 - node and edge labels and descriptions;
 - literals.
 - Semantic search based on meaning analysis:
 - vector embeddings + cosine similarity;
 - RAG (retrieval-augmented generation);
 - graph-based RAG.



SPARQL

Which works of Mozart were composed jointly with another composer?

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
```

```
PREFIX music: <http://example.com/music_graph/>
```

PROLOGUE

```
SELECT ?work ?name
```

OPERATION

```
WHERE {  
    ?wam    foaf:name      "Wolfgang Amadeus Mozart" .  
    ?work   music:composer ?wam .  
    ?work   music:composer ?other .  
    ?other  foaf:name      ?name .  
    FILTER (STR(?other) != STR(?wam)) .  
}
```

GRAPH
PATTERN

```
ORDER BY ?name
```

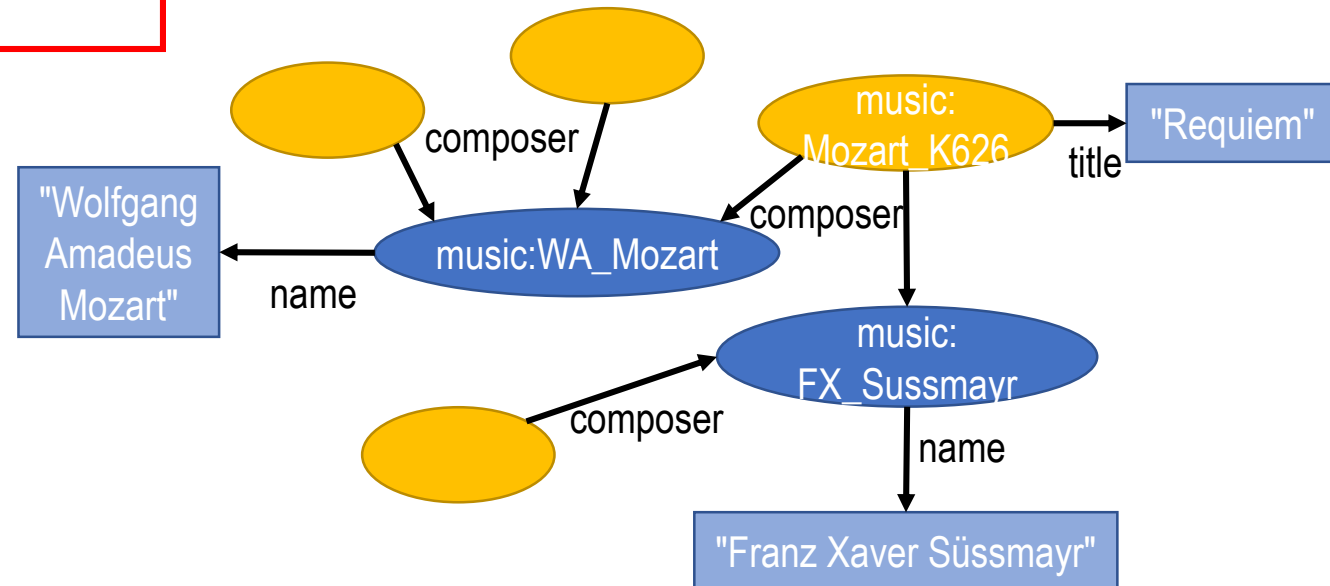
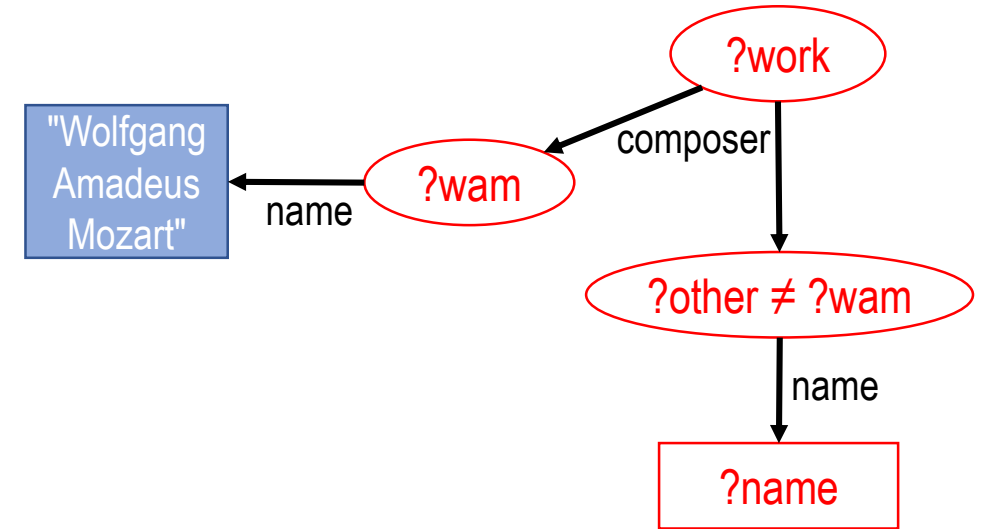
```
LIMIT 10
```

MODIFIERS

Graph Patterns

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
PREFIX music: <http://example.com/music_kg/>
SELECT ?work ?name
WHERE {
  ?wam foaf:name "Wolfgang Amadeus Mozart".
  ?work music:composer ?wam .
  ?work music:composer ?other .
  ?other foaf:name ?name .
  FILTER (STR(?other) != STR(?wam)).
}
ORDER BY ?name
LIMIT 10
```

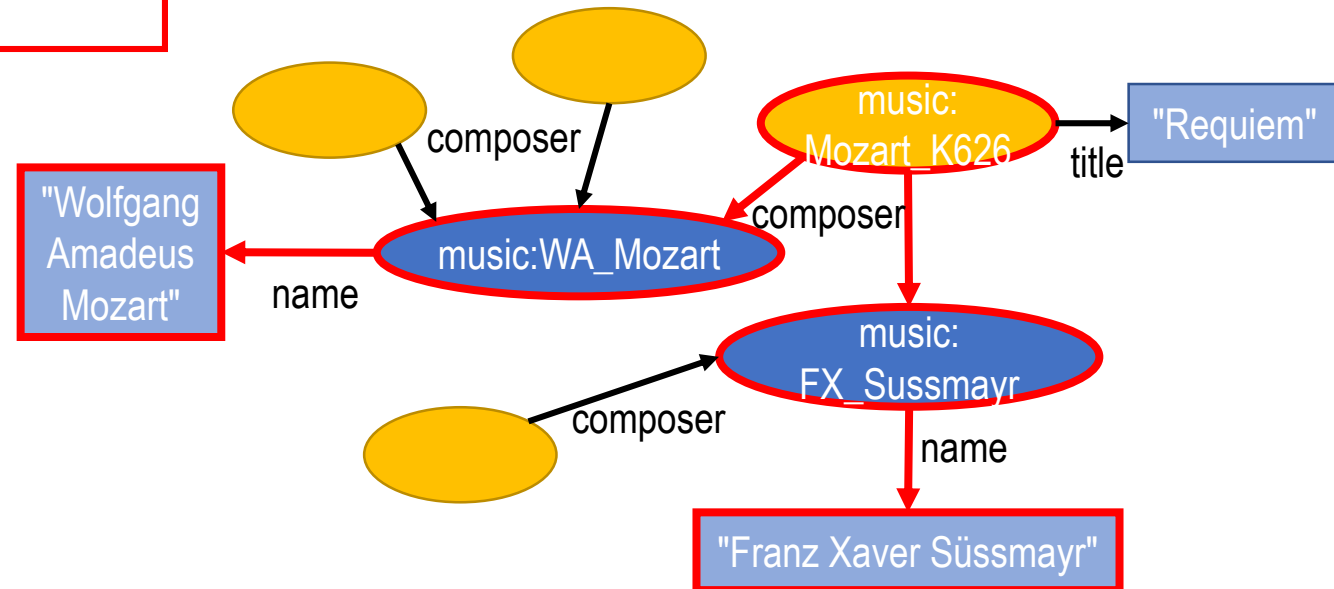
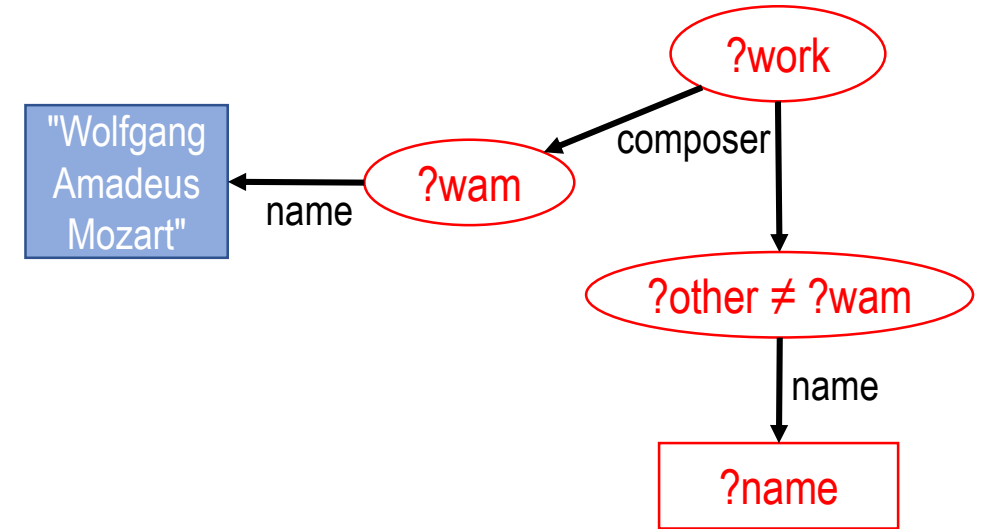
SPARQL finds results by matching the graph pattern to the target graph.



Graph Patterns

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
PREFIX music: <http://example.com/music_kg/>
SELECT ?work ?name
WHERE {
  ?wam foaf:name "Wolfgang Amadeus Mozart".
  ?work music:composer ?wam .
  ?work music:composer ?other .
  ?other foaf:name ?name .
  FILTER (STR(?other) != STR(?wam)).
}
ORDER BY ?name
LIMIT 10
```

SPARQL finds results by matching the graph pattern to the target graph.



SPARQL Operations

- **SELECT** ?v1 ... ?vn

- returns the cartesian product of possible values of variables ?v1 ... ?vn;
- as in SQL, the result can be filtered and ordered by the sequence modifiers DISTINCT, ORDER BY, OFFSET, LIMIT, GROUP BY, etc.;
- as in SQL, aggregation functions are available over variables: COUNT, MIN, MAX, AVG, GROUP_CONCAT, etc.

- **ASK**

- returns a binary result: whether the graph pattern can be found or not.

- **CONSTRUCT**

- builds a graph using values returned by pattern matching.

- **DESCRIBE**

- returns a set of triples that describe the objects matched;
- which exact set of triples depends on service configuration.

```
ASK WHERE {  
    ?x loves ?y .  
    ?y loves ?x .  
}
```

```
CONSTRUCT {  
    ?x grandmother ?z .  
} WHERE {  
    ?x mother ?y .  
    ?y mother ?z .  
}
```

```
DESCRIBE ?x WHERE {  
    ?x president France .  
}
```


SPARQL Graph Patterns

- **Basic Graph Pattern:**

- a list of triple patterns, all of which have to match;
- variables can match anything (in subject and object positions they match nodes, in predicate positions they match edges);
- in theory, the order does not matter, in practice it can affect performance.

```
WHERE {  
    ?city population ?x ;  
            ?relation Francois_Cuillandre.  
    FILTER (?x>100000 && ?x<200000)  
}
```

- **Filters:**

- **operators:** ||, &&, =, !=, <, >, <=, >=, +, -, *, /, !;
- **unary tests:** bound(?var), isURI(?var), isBlank(?var), isLiteral(?var);
- **accessors:** str(?var), lang(?var), datatype(?var), year(?date), regex(?s, "pattern").

- **Optional:**

- a triple is allowed not to match, but if it does, the variable will be filled;
- if the mayor of the city is specified, it will appear in ?mayor.

- **Union:**

- a way to express alternatives:
?city is either part of France or of Italy.

```
WHERE {  
    {  
        ?city part-of France.  
    } UNION {  
        ?city part-of Italy.  
    } OPTIONAL {  
        ?city hasMayor ?mayor.  
    }  
}
```

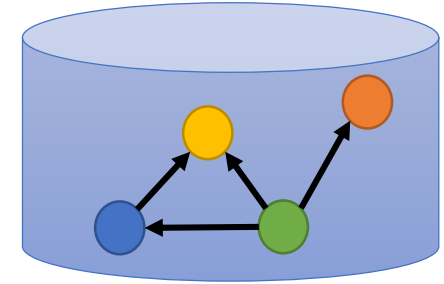
SPARQL Property Paths

- SPARQL 1.1 introduced the extremely convenient [property paths](#).
- They allow a form of reasoning over relationships.
 - **inverse:** `?a  $\wedge$ motherOf ?b` $\Rightarrow$?b is the mother of ?a
 - **chain:** `?a motherOf/parentOf ?b` $\Rightarrow$?a is the grandmother of ?b
 - **alternative:** `?a motherOf|fatherOf ?b` $\Rightarrow$?a is parent of ?b
 - **closure:** `?a childOf+ ?b` $\Rightarrow$?a is a descendant of ?b
 - you can also use `*` (zero or more), `?` (zero or one)
 - **grouping:** `?a (motherOf|fatherOf)+ ?b` $\Rightarrow$?a is an ancestor of ?b

Strengths and Weaknesses of Graph DBs

- **Strengths:**

- formal representation with heterogeneity well controlled;
- no fixed schema => extensibility at zero cost;
- reasoning ability.



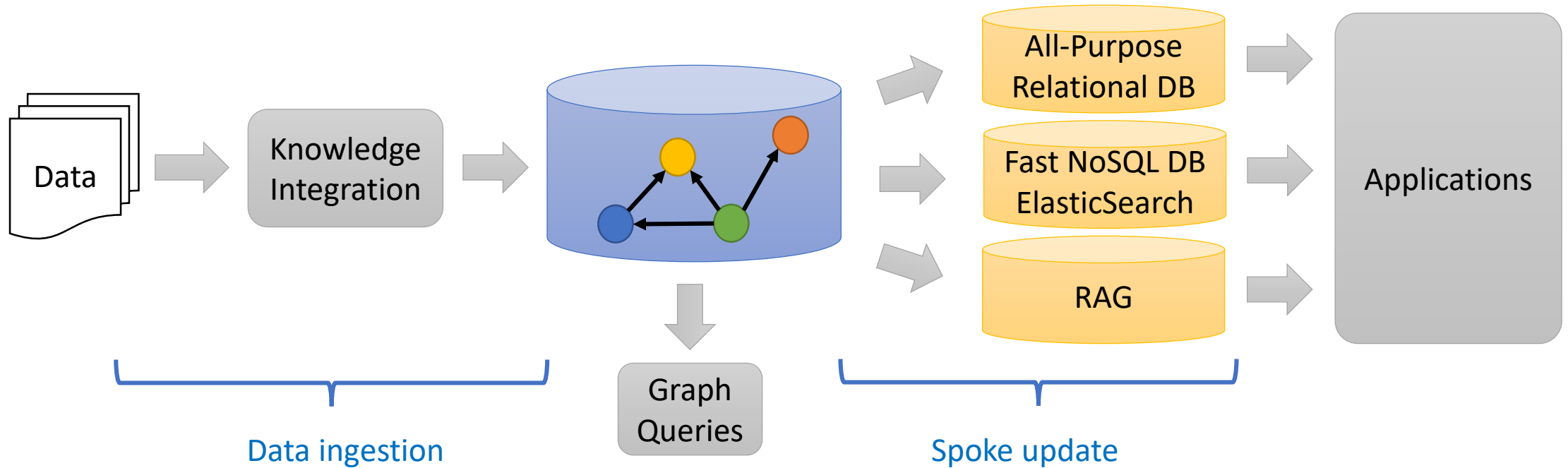
- **Weaknesses:**

- somewhat complex to query;
- slow data access, does not scale well with the quantity of data:
 - pattern matching is not easily paralellisable;
 - simple SPARQL pattern matching has a linear complexity with the graph size: $O(|P| \times |G|)$ where $|P|$ is the size of the pattern and $|G|$ is the size of the graph,
 - SPARQL with the UNION keyword is NP-complete,
 - SPARQL with the OPTIONAL keyword is PSPACE-complete (worse than NP-complete).

- **Conclusions:**

- graph DBs are rich **data hubs** for interoperability and reuse in downstream applications;
- they tend to be used as intermediary storage, in a **hub-and-spoke** architecture.

Graph DBs in a Hub-and-Spoke Architecture



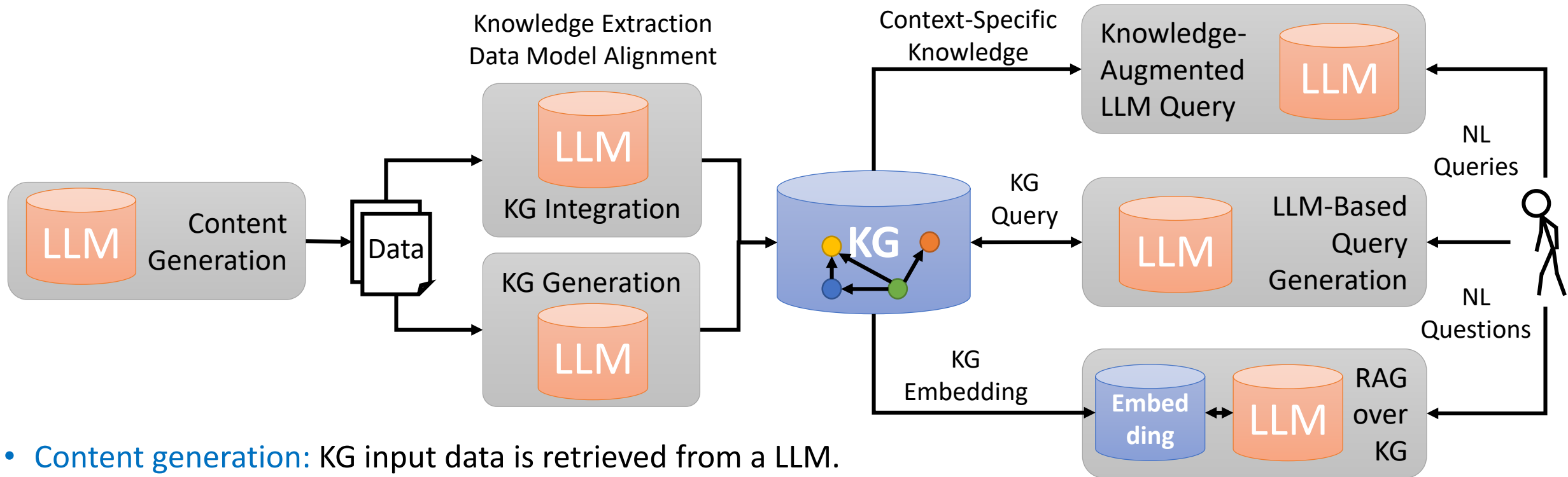
- In a hub-and-spoke architecture:
 - **hub:** the graph DB is a rich, fat, versatile, but slow central repository;
 - **spokes:** lean, fit-for-purpose DB systems optimised to downstream tasks.
- Data ingestion (i.e. knowledge integration) into the hub and export towards the spokes are separate processes.
- Only application-specific information is exported into spokes, never the full graph knowledge.
- This pattern is very common => graph DBs offer them as inbuilt processes.

Combining Knowledge Graphs and LLMs

	Strengths	Weaknesses
Knowledge Graph	dynamic content precise understandable, explainable applicable to any domain	complex to build complex query language, keyword-based search at best answer is a piece of graph reasoning limited to formal logic and graph algorithms limited generalisation ability
Large Language Model	can be queried in natural language generates answer in natural language generalises well to multiple domains multiple forms of reasoning	static content prone to hallucinations not explainable little support for long-tail and private knowledge

- The strengths and weaknesses of KGs and LLMs are complementary.
- Combining the two paradigms: one of the hottest topics of AI in the last years.
- **Warning:** much of the results in this area are still experimental, not yet ready for mission-critical applications.

Combining Knowledge Graphs and LLMs



- **Content generation:** KG input data is retrieved from a LLM.
- **KG generation:** an LLM converts input data directly into graph data, given an ontology.
- **KG integration:** an LLM assists a KG integration workflow on hard-to-automate tasks.
- **Knowledge-augmented LLM query:** the LLM answers the query but uses contextual knowledge from the KG.
- **LLM-based query generation:** from a NL question, the LLM generates a formal KG query, then transforms the KG response into a NL answer.
- **Graph RAG (Retrieval-Augmented Generation):** a KG embedding is computed, the NL query is compared to the embeddings to retrieve relevant information, a NL response is generated.

Question Answering

- Let us take a step back and consider the general problem of querying in NL.
- **Question answering:** using a background knowledge base (KG, database, etc.), produce a NL answer to a question expressed in NL.

Q: Who is the mayor of Brest?



A: The mayor of Brest is François Cuillandre.

Question Answering

François Cuillandre

3 langues

Article Discussion

Lire Modifier Modifier le code Voir l'historique Outils

François Cuillandre, né le 5 février 1955 à Brest¹, est un homme politique français, **maire de Brest** depuis 2001, et ancien député de la troisième circonscription du Finistère (de 1997 à 2002). Il est membre du Parti socialiste.

Biographie [modifier | modifier le code]

Il entre au Parti socialiste en 1977², influencé par la « deuxième gauche » de Michel Rocard³.

Après des études de droit à l'université de Brest, il devient inspecteur des impôts et est nommé à Paris. Il y suit des cours à Sciences Po Paris en cours du soir, et en 1989 soutient une thèse de droit fiscal portant sur l'harmonisation de la TVA au plan européen⁴. Il quitte l'administration fiscale en 1988 pour être chargé de mission au cabinet de Louis Le Pen⁵, alors ministre des DOM-TOM.

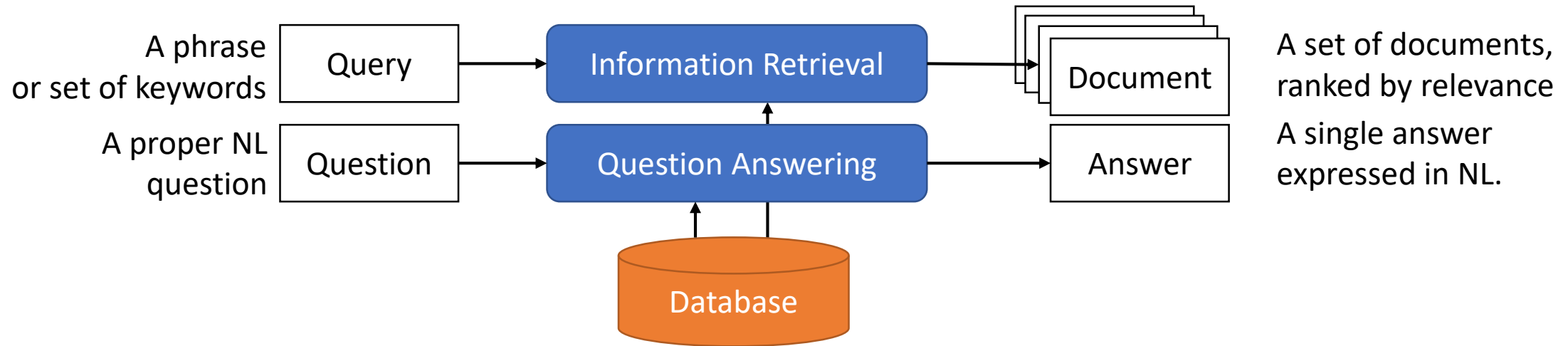
François Cuillandre



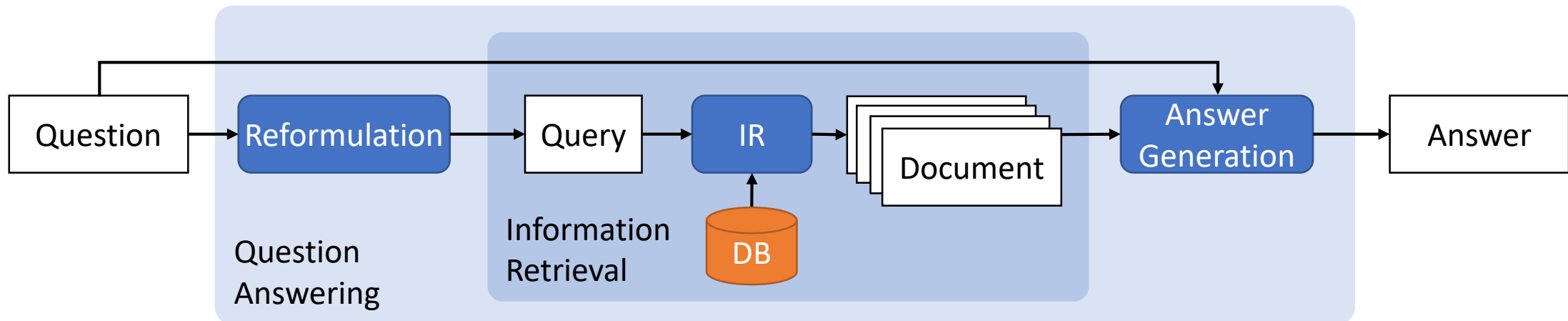
Question Answering and Information Retrieval

IR is a classic interdisciplinary problem area between NLP and information science.

QA is a strongly related problem, presupposing full, grammatical NL questions and answers.



IR is often a subcomponent of a QA system:



Retrieval-Augmented Generation for QA

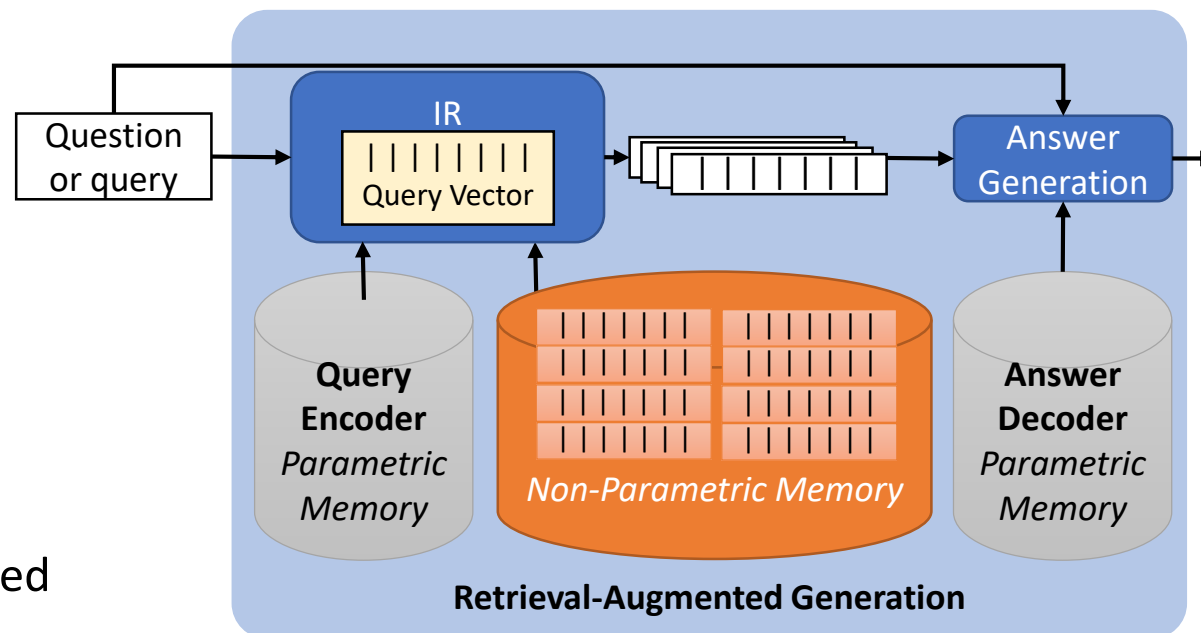
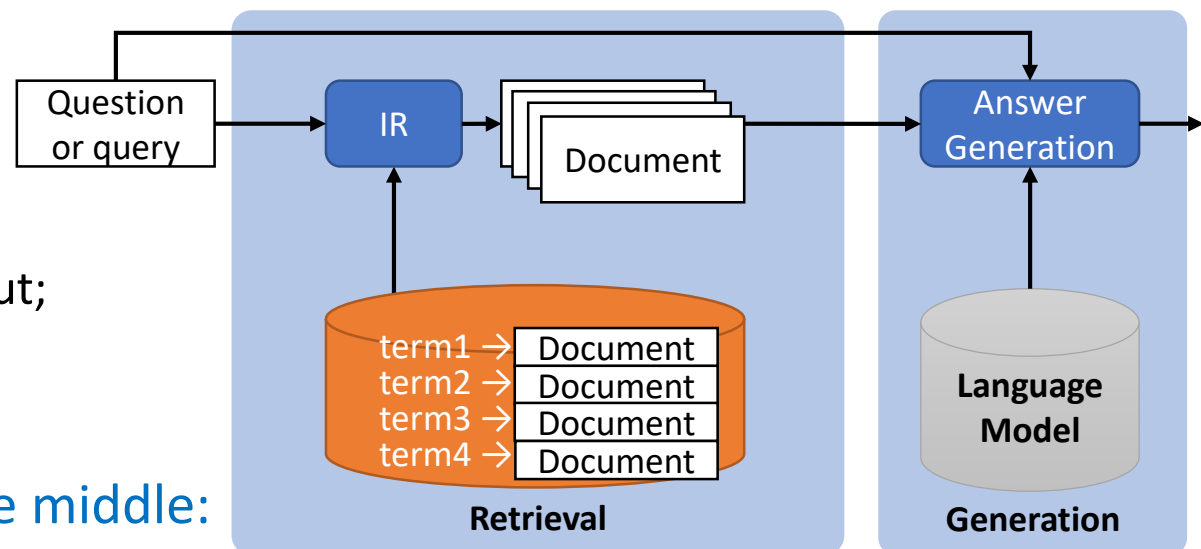
- Retrieval + generation 'pre-RAG' pipeline (not RAG, strictly speaking):

- retrieval returns actual documents;
- answer generation takes query and documents as input;
- retrieval and generation are decoupled.

- End-to-end approach: true RAG.

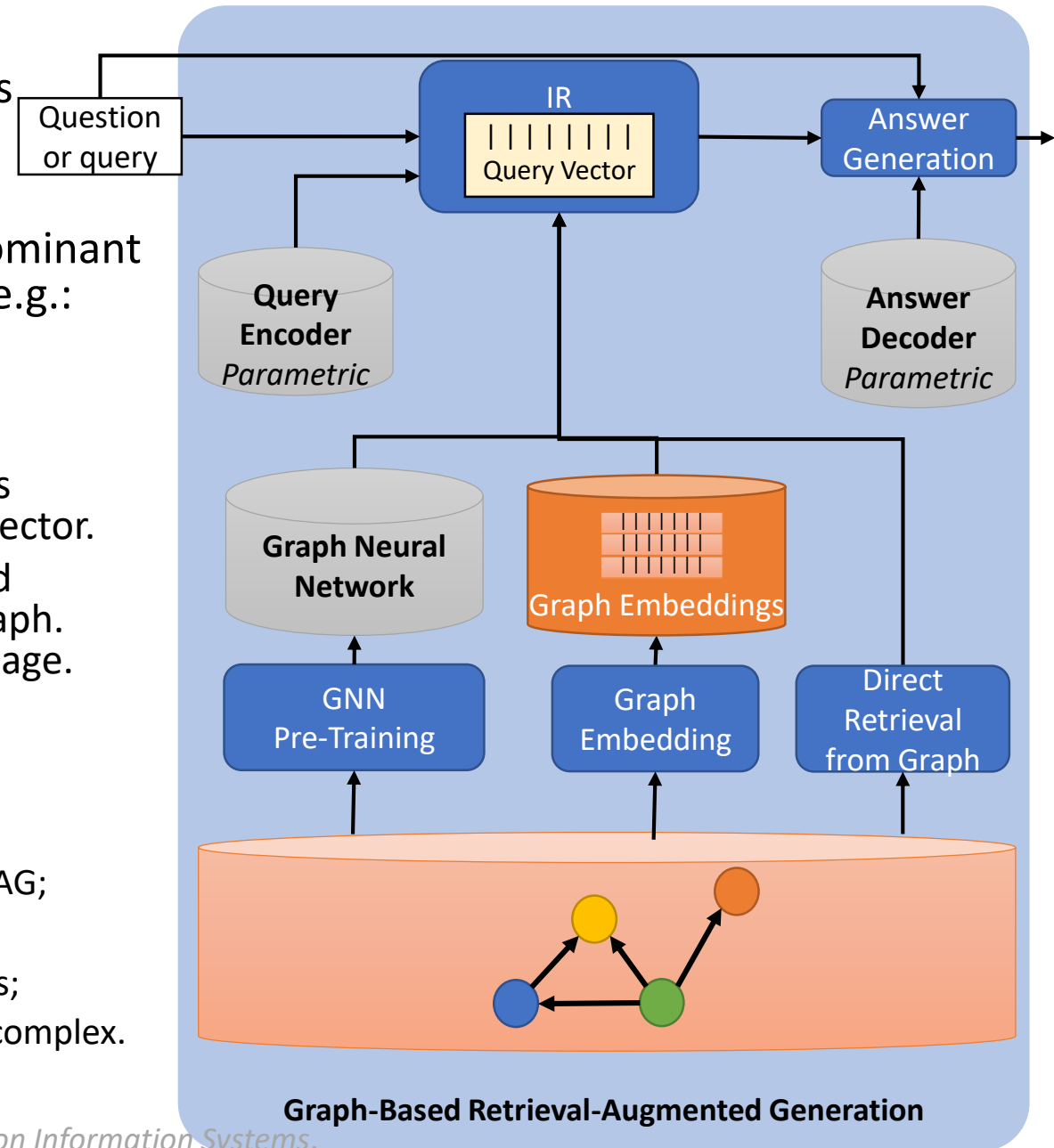
An encoder-decoder model with a vector DB in the middle:

- dense document vectors are pre-computed using a **document encoder**, e.g. BERT;
- the query is transformed into an embedding vector using a **query encoder**, e.g. BERT;
- retrieval returns dense document vectors from '**non-parametric memory**' (non-parametric: not weights of a trained DNN);
- the returned dense vectors are directly fed into the answer generator, that also receives the query and **decodes it** in a seq2seq manner;
- the query encoder and the decoder are jointly trained on query-answer text pairs.



Graph RAG

- When the data to be searched consists not of documents but of graph structures: domain knowledge graphs, social networks, etc.
- **Graph RAG** is under active research, there is no single dominant solution, companies propose their own interpretations, e.g.:
 - **Graph Neural Network**: a DNN is pre-trained to take a graph as input and to produce an output suitable to the downstream task, in our case information retrieval.
 - **Graph Embedding**: from a graph, static embedding vectors are computed, these vectors are compared to the query vector.
 - **Direct retrieval**: the natural-language query is transformed into a graph query by an LLM, executed directly on the graph. The resulting triples are then converted into natural language.
- Classic RAG vs Graph RAG: which one performs better?
 - Experiment setup in (Xiang et al, 2025):
 - a) information represented as documents, searched via RAG,
 - b) information represented as a graph, searched via Graph RAG;
 - findings:
 - classic RAG is faster and better for simple factual questions;
 - graph RAG gets better as questions grow more and more complex.

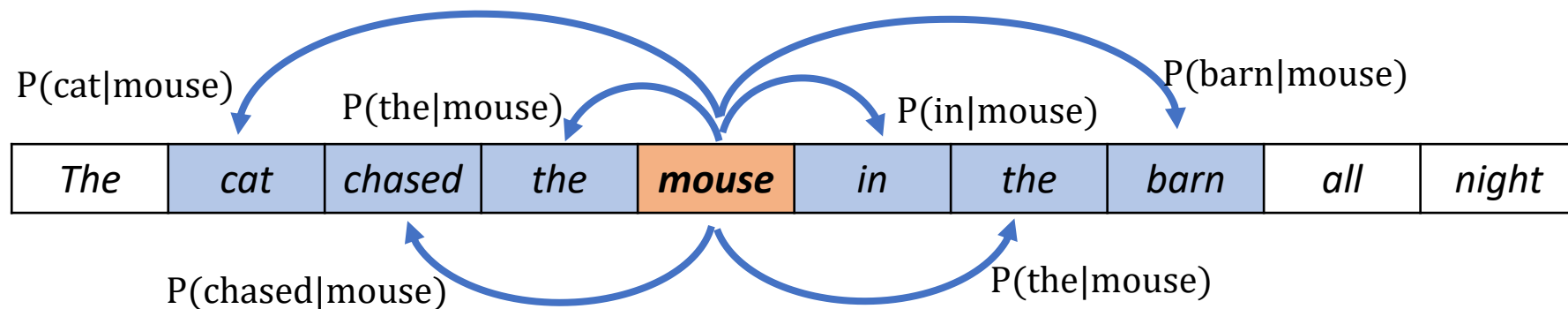


How to Embed a Knowledge Graph?

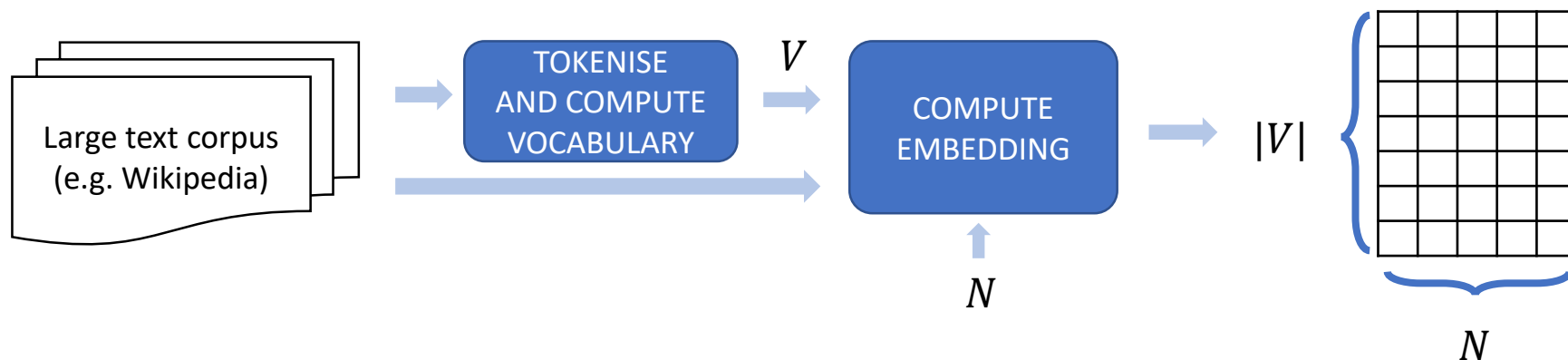
A reminder on word embeddings:

Predict a word based on its context: for all possible words w_i , compute:

$$\operatorname{argmax}_i P(w_i | c_{-n}, c_{-n+1}, \dots, c_{-1}, c_1, c_2, \dots, c_n) = P(w_i | c_{-n}) P(w_i | c_{-n+1}) \dots P(w_i | c_n).$$



- **Bag of words:** the order of words does not matter! Size of the bag = window size.
- A typical **self-supervised training** task: the *(word, context)* training pairs are generated automatically.



How to Embed a Knowledge Graph?

1. Pretend *subject–predicate–object* RDF triples in Turtle syntax are *subject–verb–object* natural-language sentences.

```
music:WA_Mozart foaf:name "Wolfgang Amadeus Mozart".  
music:Requiem_K626 music:composer music:WA_Mozart .  
music:WA_Mozart foaf:firstName "Wolfgang" .
```

2. Generate longer sentences by walking the knowledge graph (random walks for scalability).

```
music:Requiem_K626 music:composer music:WA_Mozart foaf:firstName "Wolfgang".
```

3. Replace IRIs by natural-language labels, include natural-language annotation properties.

```
Requiem composer Wolfgang Amadeus_Mozart first name Wolfgang.
```

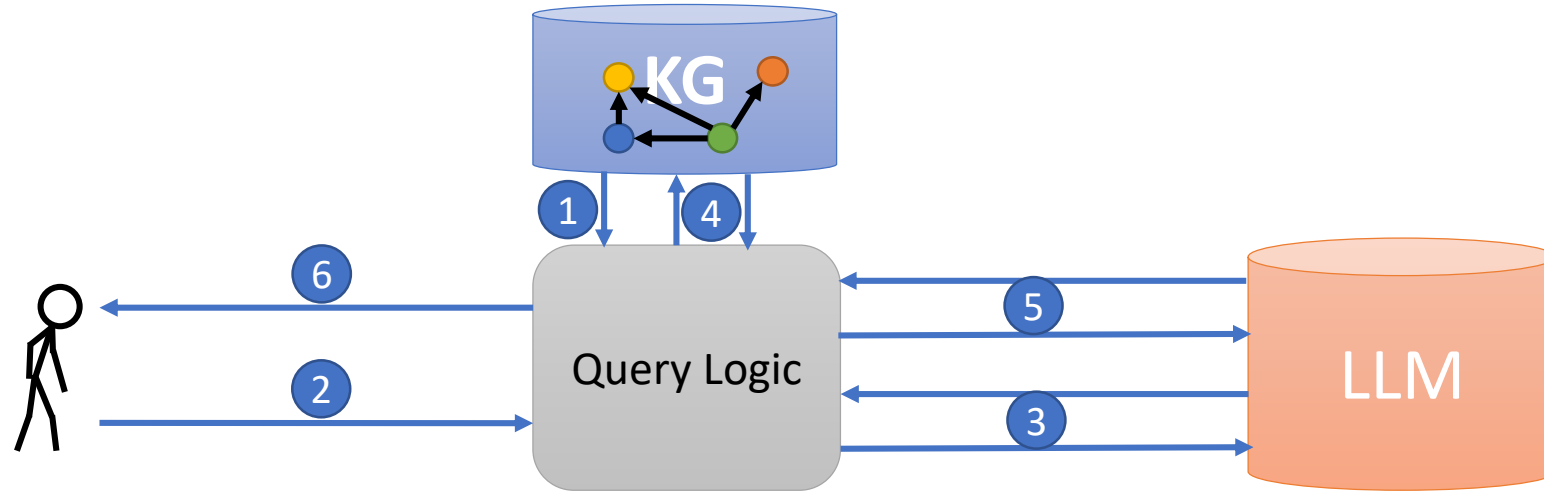
4. Train a (word or document) embedding on the corpus of sentences obtained.

Example tools: rdf2vec, owl2vec*, node2vec, graph2vec.

Example applications:

- **entity classification:** predict the most likely class(es) of an entity based on proximity in the KG (e.g. based on rdf:type);
- **entity alignment:** predict that $e_1 \equiv e_2$ based on their neighbourhood in the KG;
- **knowledge injection** into a language model.

LLM-Based Query and Response Generation



1. The Query Logic obtains the KG data model (ontology).
2. The system receives a NL query.
3. With a suitable prompt that encodes the KG *data model*, (not the entire KG which may be too big and/or confidential) an LLM is asked to convert the NL query into a SPARQL query.
4. The resulting SPARQL query is sent to the graph DB.
5. With another suitable prompt, the Query Logic asks the LLM to convert the graph DB's response into a natural-language response.
6. The NL response is sent back to the user.

What is the birthdate of Mozart?

```
SELECT ?bdate WHERE {  
  ?x foaf:surname "Mozart";  
  bdo:birthDate ?bdate.  
}
```

1756-01-27

Mozart was born on 27 January 1756.

Today's Lab

https://github.com/gbella/NLP/raw/refs/heads/main/Knowledge_Processing/KG_PhD_lab_resources.zip